Does the Florida Man have the 'most correct' name? - Identifying and Exploring Racial and Linguistic Bias in Spell Checker Tools

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Abstract

This study investigates whether disparities in spell-checking systems' treatment of personal names reflect isolated system behaviour or broader structural bias. It develops a large-scale evaluation framework to analyse recognition and correction behaviour across multiple spell-checking tools. The analysis combines script-level comparisons, regional and ethnicolr-based aggregation, and controlled regression to isolate potential sources of variation. Results show systematic differences in recognition rates across origin groups, with disparities persisting under Latin-script conditions and after controlling for key confounding factors. This suggests that observed differences are not solely attributable to script variation or orthographic complexity. Analysis of correction behaviour further indicates that unrecognised names are more likely to be replaced with forms associated with more frequently represented European and North American naming conventions. Overall, the findings suggest that these disparities are not tool-specific, but reflect broader imbalances in lexical coverage and representation. The study therefore positions equitable name recognition as a core design requirement in language technologies, rather than a secondary or edge-case concern.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics committee.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(*Saraswati Brown*)

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Chapter 1

Introduction

1.1 Research Motivation

Spell-checking tools are among the most widely and silently deployed language technologies in digital life. Embedded in word processors, web browsers, and communication platforms, they operate with minimal user scrutiny while continuously making judgements about what constitutes "correct" written language.

Despite their prevalence, their treatment of personal names from racially and linguistically diverse backgrounds is largely overlooked in academic and public discourse. Names frequently fall outside standard morphological rules and common lexicons, meaning their recognition depends heavily on the scope and composition of a tool's dictionaries. As a result, names that do not conform to dominant linguistic norms or are underrepresented in the corpora from which the dictionaries are derived may be systematically flagged as incorrect. This raises the possibility that spell-checking systems do not merely reflect linguistic variation, but actively reinforce existing cultural and linguistic hierarchies.

Existing research shows that this is not solely a theoretical concern, but a practical and recurring experience for users which reflects structural limitations of current approaches to text correction. Deshmukh et al. [1] demonstrate that traditional spell-checking systems, particularly those based on finite automata and dictionary lookup, struggle with proper nouns that fall outside their predefined lexical reference frames. In a broader context, Dyal-Chand [2] argues that software systems built on culturally homogeneous datasets risk reinforcing exclusionary norms. This critique directly applies to the hypothesis of the limitations of English-dominant lexicons that underpin many spell-checking tools.

Beyond academic research, public awareness campaigns such as #IamNotATypo highlight the lived experience of this issue, drawing attention to the repeated flagging and correction of non-European/Latin-origin names in everyday digital contexts. These accounts suggest that the limitations identified in formal research manifest as persistent user-facing friction. This signals both a lived-experience problem and a gap in technical accountability.

This project responds to that gap by developing a large-scale empirical framework for evaluating how spell-checking tools treat names from diverse linguistic and cultural backgrounds. By analysing recognition and correction behaviour across multiple tools and a globally representative dataset, it aims to determine whether observed disparities are isolated to specific systems or indicative of broader, systemic bias. In doing so, this work reframes spell-checking systems not as neutral assistive tools, but as systems where linguistic norms are encoded, enforced, and potentially contested.

1.2 Research Questions

This project investigates whether spell-checking tools are systematically biased against certain linguistic and cultural backgrounds. The central research question is:

Are spell-checking tools systemically biased against names from Global Majority origins?

To address this, the study is structured around the following research questions:

1. **Detection of disparity:** Do spell-checking tools exhibit systematic differences in recognition rates between Eurocentric and Global Majority naming conventions?
2. **Mechanism of bias:** To what extent are observed disparities driven by script-level differences versus lexical limitations within a shared writing system?
3. **Correction behaviour:** When names are flagged as incorrect, what patterns emerge in the corrections offered, and do these systematically reflect preference for Eurocentric naming conventions?

Together, these questions identify whether disparities exist, explain their underlying causes, and finally to examine their practical consequences within user-facing systems.

1.3 Contributions

This project makes the following contributions:

- **A globally diverse, enriched name dataset for bias analysis:** The construction and enrichment of a large-scale dataset of approximately 727,000 names, incorporating script classification, transliterated forms, and predicted cultural and geographic origin. This dataset provides a structured resource for investigating linguistic diversity and representation in computational language systems.
- **A scalable, multi-tool evaluation framework:** The development of an automated pipeline that systematically evaluates name recognition and correction behaviour across multiple spell-checking architectures under controlled experimental conditions. By applying each tool to both original and transliterated inputs, the framework enables a consistent large-scale comparison across heterogeneous systems.

- **A methodological approach for isolating sources of bias:** The introduction of a dual-condition experimental design that distinguishes between script-level limitations and lexical bias within each system, providing a way to attribute observed disparities to specific underlying mechanisms. This distinction is not consistently made in prior work.
- **Empirical evidence of systemic bias in name recognition and correction:** A comprehensive quantitative analysis demonstrating that disparities in recognition rates persist across multiple spell-checking tools, indicating systemic bias rather than tool-specific artefacts. The study further shows that correction suggestions disproportionately favour names from European-centralised origins, revealing how bias extends beyond detection into user-facing behaviour.

Chapter 2

Background

2.1 Overview

Spell checking tools are a core component found in most text-based digital platforms. Their apparent function, to detect potential orthographic errors and provide corrections, seems simple. But this does not reflect the extent to which these systems embed a series of design decisions about what constitutes valid language and which linguistic forms are represented and which are excluded. In enforcing linguistic 'correctness', they encode implicit assumptions about which language varieties, orthographies, and naming conventions are deemed legitimate.

It follows that such assumptions may reinforce a linguistic hierarchy that supports structures of marginalisation and exclusion of certain linguistic and cultural communities. This project takes the position that these assumptions are structurally embedded rather than incidental, and therefore require empirical investigation at scale rather than anecdotal observation. In this section, we explore the functionality of error detection methods employed in common spell-checkers and consider the broader implications of their limitations.

2.2 How do Current Spell-Checkers Detect Errors?

The first spell-checking models existed as stand-alone packages focused only on detecting orthographic deviations, acting as verifiers rather than providing guidance or correction[3]. As these programs became integrated into operating systems and word processors, spell-checkers transformed into interactive correction systems, capable not only of detecting misspellings but also of offering suggestions and grammatical feedback.

Currently, spell checkers work by comparing user inputs to a reference lexicon and then applying a series of algorithms to propose potential corrections. This is largely done by two main methods, morphological decision making or algorithmic rule-based systems, and now, with the rise of Artificial Intelligence (AI), also machine learning and AI-based models.

2.2.1 Morphological Systems

In these systems, each word's correctness is defined explicitly by whether it appears in a dictionary or can be formed through valid morphological transformations. This method focuses on rule-based linguistic design of affixes, stems and morphological compounding.

When a user input is processed as a token, the system checks whether it exists in the dictionary or can be generated through the rules encoded in the affix file. For example, from the root 'form', the affix rules could generate 'forming' and 'formed'.

Hunspell[4], one of the most widely deployed morphological spell-checkers (used in applications such as LibreOffice, Mozilla Firefox and Google Chrome), exemplifies this approach. Its dictionary is supplemented by affix rules that allow the generation of valid word forms from a base lexicon, supporting high precision in morphologically rich languages such as Hungarian or Finnish. However, whilst this approach supports high precision handling of inflectional variation, it remains dependent on the presence of base forms within the dictionary. Personal names, which are typically stored as stand alone lexical entries rather than generated forms, must be explicitly included to be recognised. Names from underrepresented linguistic or cultural groups are less likely to appear in the dictionary and cannot be produced through morphological derivation, creating systematic disparities in recognition and revealing structural vulnerabilities when encountering linguistic forms outside predefined rules[5].

2.2.2 Algorithmic and Probabilistic Systems

Both algorithmic and morphological systems rely on a dictionary to evaluate the correctness of user input. Unlike morphological systems which have an understanding of linguistics, algorithmic systems treat words as symbolic strings or tokens, applying computational distance measures to estimate the proximity of an unrecognised word, to entries in the dictionary.

The most common measure is the Levenshtein distance, which calculates the minimum number of insertions, deletions, or substitutions required to transform one word into another [6]. This method prioritises orthographic similarity ('closeness') in spelling rather than grammatical context, reducing names to sequences of characters evaluated through distance metrics with lack of consideration for linguistic and cultural validity. There are a variety of similarity algorithms which offer different ways to quantify closeness between strings, for example, the Hamming distance measures the number of positions at which two strings of equal length differ. Jaro-Winkler similarity gives higher weight to characters that match near the beginning of a word, making it effective for typographical errors in short names or common terms [7]. N-gram models compare overlapping sequences of characters to capture partial matches and phonetic resemblance. These alternative metrics often perform better for certain languages or error types, though they share the main limitation of abstracting words into tokens and studying numerical relationships rather than linguistic context. This allows them to misinterpret names or words that deviate from dominant spelling conventions[8]. These systems are typically paired with frequency-ranked dictionaries, meaning that recognition and correction are

not only functions of string similarity but also rely heavily on the composition of the underlying corpus.

SymSpell[9] exemplifies this design by combining edit-distance-based candidate generation with a frequency dictionary derived from the intersection of large-scale corpora (Google Books Ngram data) and curated lexical resources (SCOWL - Spell Checker Oriented Word Lists). This design prioritises high-frequency, attested vocabulary while filtering out rare or noisy entries. While this method is effective for general spelling correction, it introduces a structural bias rooted in frequency-based inclusion. Names that are less frequent in these corpora or that fall outside standard English vocabulary, are unlikely to be included in the dictionary and therefore cannot be recognised or suggested as corrections. As a result, linguistically dominant forms are systematically favoured, not through explicit design intent, but due to the statistical properties of its underlying data sources.

LanguageTool[10] is a hybrid system that combines rule-based grammar checks with dictionary lookup, and in some configurations uses Hunspell dictionaries internally for spelling. Its context-aware rules allow it to detect and sometimes correct errors that purely word-level spell-checkers might miss. However, like other dictionary-driven systems, it remains constrained by the coverage of its underlying lexicons. Low-frequency names absent from these dictionaries are often unrecognised, and context-sensitive rules cannot generate valid forms for them. Additionally, LanguageTool's handling of case and script can result in non-Latin inputs being passed through without correction, creating a measurement artefact by inflating apparent recognition rates without reflecting true lexical familiarity. This illustrates that hybrid systems may offer broader functionality, but still reproduce underlying biases present in their dictionaries.

2.3 Ethical and Social Implications

Spell-checking tools are not neutral technologies. Although they are designed to standardise written language and assist users, the assumptions embedded into their design can reinforce wider social and cultural hierarchies. When certain names or linguistic forms are persistently flagged as incorrect, this does not simply reflect a computational limitation, but shows how linguistic conventions uphold social power [11]. The following sections outline how these technical mechanisms intersect with identity, representation, and fairness in digital language systems.

2.3.1 Names and Identity

Personal names have significant linguistic and cultural meaning, acting as markers of identity, heritage, and social belonging. They act not only as identifiers, but also as carriers of cultural and historical context[12]. However, within computational systems names are often abstracted to tokens of data to be matched, verified, or corrected. This abstraction strips names of their depth, resulting in a technological interpretation of identity that privileges standardisation over diversity.

Both medical and sociological research demonstrates that the modification or misrep-

resentation of names can have psychological and social consequences [13][14]. In technological contexts, this misrepresentation manifests when digital systems, such as spell-checkers, treat unfamiliar names as invalid. Such processes implicitly reinforce dominant linguistic norms, defining certain identities as 'incorrect' within computational space.

This is compounded by the structure of many language technologies, which depend on lexicons that are largely monolingual and European[11]. Consequently, names of African, Asian, or Indigenous origin are disproportionately excluded from the reference systems used by mainstream spell-checking tools. This results in a persistent cultural asymmetry.

2.3.2 Bias Mechanisms and Algorithmic Erasure

Bias in computational systems often occurs due to structural and statistical features of the data and algorithms that underpin them, rather than solely from overt design decisions. Three main mechanisms through which linguistic and demographic biases emerge in natural language technologies are underrepresentation in training or reference data, algorithmic abstraction that simplifies language into numerical relationships, and feedback loops that reinforce majority linguistic norms[15, 16, 11]. These factors collectively determine which linguistic forms are recognised as valid and which are marginalised.

In spell-checking systems, these mechanisms manifest through the construction of dictionaries, the weighting of frequency information, and the selection of similarity metrics, which collectively shape which names are recognised, rejected, or replaced. The reliance on inherently limited lexicons means that culturally specific names are often absent from the dictionary entirely. When an unrecognised name is flagged, the system generates replacement suggestions based on similarity or frequency metrics. This process, while appearing computationally neutral, reinforces social asymmetry by validating names that conform to dominant linguistic norms, while those that do not are repeatedly highlighted as 'invalid' or replaced by more familiar forms.

Caliskan et al.[15] showed that word embeddings trained on large text corpora replicate social stereotypes seen in human behaviour, while Swinger et al.[16] found that even after debiasing, residual associations persist across gender and ethnicity. Similarly, Berlincioni et al.[17] observed large performance gaps for African or African-American names in multimodal large language models. These findings uphold the suggestion that linguistic inequities are systemic, arising from the datasets and representations that inform algorithmic decisions rather than from isolated design flaws.

Within spell-checking systems, these inequities can result in 'algorithmic erasure', the systematic omission or correction of names that fall outside of a system's learned norms. Each instance of correction may appear minor, but collectively they contribute to a pattern of digital exclusion, in which certain linguistic identities are rendered invisible within computational environments. As Qadri et al.[18] note, large-scale language models risk amplifying this erasure by embedding bias within their probabilistic structures, making it increasingly difficult to trace or correct. This highlights the need for critical

evaluation of the datasets and decision rules that shape everyday language technologies.

A further complication noted in critical literature is that algorithmic inclusion is not wholly beneficial. Although exclusion from computational systems constitutes a form of symbolic and functional erasure, increased legibility may also expand the scope of classification and surveillance. Benjamin [19] explores the idea that visibility within algorithmic systems can operate as both a site of recognition and a mechanism of control, under what she terms the “New Jim Code”, in which automated systems, often perceived as neutral, reproduce or deepen social inequities by making marginalised groups visible to govern and control them. In this sense, the mechanisms that produce algorithmic erasure operate in tension with concerns over visibility and governance. For groups already subject to disproportionate monitoring, partial opacity to automated systems may carry a limited protective dimension, even as it simultaneously reflects exclusion from linguistic and computational recognition.

2.3.3 Standard Language Ideology

Linguists use the term *standard language ideology* to describe the belief that one variety of a language is inherently correct, superior, or more legitimate than others. It is widely argued that “standard” language is not a neutral linguistic fact, but a socially constructed framework that privileges particular forms while marginalising variation [20, 21]. What is considered “standard” is therefore not simply a matter of grammatical correctness, but a reflection of historical, political, and cultural power.

In digital systems, this ideology is operationalised through design choices. Decisions such as which dictionary is bundled, which language varieties are supported, which names are included, and what constitutes an acceptable error rate all contribute to defining the boundaries of "correct" language. Spell-checkers are computational embodiments of this ideology; they render the "standard" legible and systematically mark deviations as errors.

Phillipson’s concept of *linguistic imperialism* [22] extends this argument to a global scale, highlighting how the dominance of English-language technologies reinforces the authority of particular linguistic norms across multilingual contexts. As many widely used spell-checking tools are built on English-dominant corpora and lexicons, they implicitly prioritise Western orthographic conventions and naming systems. This results in a form of structural asymmetry, where names and language varieties from the Global Majority are more likely to be excluded, flagged, or corrected, not due to inherent linguistic deficiency, but due to their systematic absence from the datasets and lexicons that define correctness.

Taken together, these perspectives frame spell-checking not simply as a technical task, but as a mechanism through which linguistic norms are enforced and reproduced. This provides a theoretical foundation for examining whether such systems systematically disadvantage certain cultural and linguistic identities, as explored in this project.

2.4 Situating This Project Within Prior Bias Literature

Prior work has established the use of personal names as effective probes for detecting algorithmic bias. Caliskan et al. [15] demonstrated that name lists can function as demographic proxies, revealing latent social stereotypes in word embeddings. This approach was further extended by Swinger et al. [16] who applied similar techniques to evaluate fairness in Natural Language Processing. Together, these studies provide a methodological foundation for using names as indicators of demographic representation.

More directly related work has examined bias in systems that operate explicitly on names. Karakasidis & Pitoura [8] identify bias specifically in name matching tasks, demonstrating that even similarity-based systems can exhibit uneven performance across demographic groups. Hafner et al. [11] extend this through an audit of name-ethnicity classification systems, highlighting measurable performance gaps across different name origins. Their work is particularly relevant in both scope and methodology, as it evaluates bias using large-scale name datasets and supports the use of tools such as Ethnicolr for demographic inference.

Beyond computational systems, Bertrand & Mullainathan [23] establish real-world context as name-based discrimination is already present in hiring before any digital system is involved. This establishes that any bias observed in spell-checking systems does not emerge in isolation, but instead has the potential to compound existing patterns of social disadvantage.

Blodgett et al [21] provide a critical framework for evaluating bias research in natural language processing, arguing that prior work frequently fails to ground analysis outside the field, state explicitly who is harmed, or engage with lived experience. This project adopts these critiques as a core structuring framework. First, the analysis is grounded in interdisciplinary literature spanning sociolinguistics, identity research, and studies of discrimination, treating representational harm as significant in its own right rather than as a purely technical issue. Second, the harms under investigation are explicitly defined: disparities in spell-checking disproportionately affect names from Global Majority linguistic traditions, occurring silently within widely used digital systems. Third, while direct engagement with affected communities is beyond the scope of this computational study, initiatives such as #IamNotATypo and related sociological research are incorporated as forms of lived-experience evidence. Future work is proposed to extend this analysis through community-centred methodologies, including user studies and accessible evaluation tools. A further tension highlighted by Blodgett et al. is the "categorisation paradox," whereby the use of demographic categories to measure bias can inadvertently reinforce the very distinctions it seeks to critique. This limitation is acknowledged within the design of this study and is discussed further in Section 3.5.

Chapter 3

Methodology

3.1 Dataset

The dataset used in this study was constructed using the `names_dataset` Python library[24], which provides aggregated name-country frequency distributions derived from large-scale social network data. The underlying data is derived from a publicly released dataset of approximately 533 million Facebook users from 2021. According to the library documentation, this resource contains over 730,000 unique first names and 983,000 surnames, each associated with a probabilistic distribution across countries based on the observed locations of users.

Each name is associated with a country-level probability distribution, from which the most likely country of origin (`top_country`) and its corresponding probability (`top_country_prob`) were derived. These probabilities reflect relative frequency within the Facebook user base rather than true population distributions, but provide a large-scale proxy for geographic name usage across 105 countries.

A series of cleaning steps were applied to ensure the dataset represented meaningful inputs for spell-checking systems. Names consisting solely of single-character tokens (e.g. "A A") were removed, as these are likely artefacts of user metadata rather than genuine names, and therefore do not provide meaningful signals for origin classification. Multi-word names were retained and treated as single tokens, reflecting how real-world spell-checkers process whitespace-delimited input.

Further cleaning was performed after the corrections pipeline was run. This included removing whitespace and noisy characters such as `*_:, 0-9` but retaining common naming punctuation such as hyphens and apostrophes (e.g. Jean-Claude, O'Brian) which are recognised in onomastic research as legitimate elements of personal names[25]. While real-world naming practices can include a wide range of characters that fall outside conventional programming assumptions, spell-checking systems themselves operate over these constrained orthographic expectations and therefore cannot reliably process digits and non-standard punctuation, so names with such characters would be uniformly flagged irrespective of linguistic origin. Thus, their inclusion would introduce noise to the evaluation rather than informative variation. By performing this cleaning

post-correction, we were able to evaluate whether these steps significantly impacted flagging rates and justify the choice to apply them.

The use of this dataset introduces several important limitations. As the data is sourced from Facebook, it reflects the demographic distribution of the platform's user base rather than the global population. Certain regions may be under-represented due to limited Facebook access, including countries such as China, Iran, and North Korea, where access is restricted as part of broader state censorship and information control policies[26]. Additionally, while the dataset captures everyday name usage across a broad range of users, it does not guarantee accurate identity information, as names and locations are self-reported. As there is no strict definition of a personal name, entries such as brand names or community groups may be incorrectly included. These were retained as their sparse occurrence is unlikely to materially affect aggregated rates by country, region, or ethnic group.

Despite these limitations, the dataset offers several advantages over alternative sources. Its scale and geographic breadth significantly exceed commonly used datasets such as Wikipedia-derived name lists, which are biased towards notable individuals and editorial selection processes. By contrast, the Facebook-derived dataset captures a wider range of everyday naming practices across diverse socioeconomic contexts. Alternative approaches, such as using national census data, were considered but ultimately not adopted. While census data provides high-quality, country-specific name distributions, it would be crucial to combine many national datasets to achieve comparable global coverage, which would introduce additional complexity and potential inconsistencies across reporting standards and time periods.

The use of this dataset also raises ethical considerations. The original Facebook data was obtained and released without the consent of the individuals involved[27]. However, this study operates exclusively on aggregated name-frequency distributions and does not include any personally identifiable information. No attempt is made to analyse or infer characteristics at the individual level, and no user-specific data is accessible within the dataset. Furthermore, the data is already publicly available and widely used in research contexts, meaning that this work does not extend or amplify the original breach.

3.2 Phase 1: Dataset Enrichment

To enable analysis of script-level and lexical biases, the dataset was enriched with additional annotations. This enrichment involved script detection, transliteration, extended probabilistic origin classification, language identification, and cross-validation against country-specific linguistic data.

Names were first assigned a script category (`name_script`) by extracting the script prefix from the Unicode character name (e.g. `CJK` from `CJK UNIFIED IDEOGRAPH-4E2D`), covering Latin, Arabic, Cyrillic, Devanagari, Hebrew, CJK, and others. This classification provided the basis of distinguishing errors arising from script encoding from those arising from lexical content. Non-Latin scripts were transliterated to Latin characters using the `anyascii` Python library[28], producing a `name_latin` field. Latin-script

names were left unchanged. Transliteration introduces limitations: for example, Arabic abjad transliterations lose vowel information (محمد → "mhmd" rather than "Muhammad"), which may overstate lexical bias for Arabic-origin names. Vietnamese diacritics were stripped ("Nguyn" → "Nguyen"), which is less severe but still phonemically meaningful. Global Majority Latin-script names, such as many Nigerian, South Asian, and Vietnamese names, were unaffected by transliteration, providing the clearest evidence for lexical bias in these subsets.

Each name was then processed with `ethnicolr`, a character-level classifier trained on a combination of Wikipedia data, US census data and Florida voter registration data [29], predicting 13 origin categories (e.g., British, East Asian, African, Muslim, Indian Subcontinent). Outputs include the top category (`ethnicolr_race`), the associated confidence (`ethnicolr_prob`), and the full probability distribution (`eth_distribution`). A two-pass strategy was employed: names were first classified in their original form, and any NaN returns were reprocessed using `name_latin`, affecting approximately 18% of the dataset and primarily non-Latin-script names. A limitation is that the model is trained on Wikipedia, over-representing notable individuals and potentially reducing its precision across the broader Facebook-derived dataset.

In order to provide an additional validation signal for the frequency-based origin probabilities defined by the dataset, character-level language identification was performed using the `langdetect` library[30] to populate the `langdetect_lang` field with detected language codes, language names in (`langdetect_lang_name`), and confidence scores in (`langdetect_prob`). Short single-token names produce unreliable detections; to mitigate this, each name was repeated five times prior to detection (e.g., "Omar Omar Omar Omar Omar"). A boolean field (`country_lang_comp`) indicates whether the detected language matches any of the official languages of the name's most probable country (`top_country`), providing a cross-validation indicator for annotation quality.

Official languages for each top country were retrieved using the `countryinfo` library and stored as `top_country_langs` (ISO 639-1 codes), enabling systematic evaluation of detected language alignment with expected country-level linguistic contexts.

Finally, an attempt was made to query the Nationalize.io API for a stratified random sample of 1,000 names, divided into quartiles by `top_country_prob`. Due to API rate limits and very low coverage (<0.2% of the dataset), these results were not incorporated into the primary analysis.

This enrichment phase provided a multi-dimensional framework for analysing script, lexical, and language-origin biases, while explicitly accounting for limitations introduced by transliteration, classifier training data, and short-token noise.

3.3 Phase 2: Corrections

3.3.1 Dual-condition experimental design

To isolate different sources of bias, a dual-condition design was implemented for the testing framework. In Condition A (*Original*), each name was passed to the spell-

checking systems in its original form, preserving its native script. In Condition B (*Latin*), the transliterated form (`name_latin`) was used, ensuring that all inputs were represented in ASCII and therefore comparable across tools regardless of script support.

The difference in recognition rates between Condition A and Condition B provides a measure of script-level bias, capturing the extent to which each system is technically limited in its ability to process non-Latin scripts. Within Condition B, where all names share a common script, disparities in recognition across origin groups can be attributed to lexical bias, reflecting differences in dictionary coverage and underlying data distributions. This distinction is critical, as the aim of this study is not merely to demonstrate the known technical scope of spell-checking systems, but to examine whether disparities persist beyond script constraints. By separating script-level effects from lexical behaviour, the design enables a more precise evaluation of lexical bias within a shared representational space.

For names that are already in Latin script but originate from Global Majority linguistic contexts (e.g. many Nigerian, South Asian, and Vietnamese names), Conditions A and B are identical. In these cases, any observed disparity is entirely lexical in nature, providing the clearest evidence for the central research question.

3.3.2 Tool Selection & Reasoning

Three spell-checking tools were selected to represent distinct architectural approaches: `Hunspell`[4], `SymSpell`[9], and `LanguageTool`[10]. By evaluating systems with different underlying mechanisms, the study aims to determine whether observed disparities are tool-specific or indicative of broader structural patterns.

`Hunspell` is a widely deployed morphological spell-checker that combines dictionary lookup with affix-based rules to generate valid word forms. Its use in applications such as LibreOffice, Mozilla Firefox, and Google Chrome makes it representative of rule-based systems operating at scale in real-world environments. As such, its behaviour is particularly important to evaluate, as any systematic limitations directly affect a large global user base.

`SymSpell` implements a purely algorithmic approach based on the symmetric delete algorithm, using precomputed edit-distance candidates and frequency-ranked dictionaries for efficient lookup. Unlike morphological systems, it does not incorporate linguistic rules, instead treating words as character sequences evaluated through similarity and frequency. This abstraction does not provide special handling of personal names, making it a useful test case for assessing how string-based systems respond to linguistically diverse inputs.

`LanguageTool` is a hybrid system that integrates dictionary-based spell-checking with rule-based grammatical analysis. It operates at both the word and sentence level, enabling context-aware corrections in addition to standard spelling detection. In some configurations, it relies on Hunspell dictionaries for lexical validation, but extends this with additional rule-based processing.

Each of the tools carried unique differences which required careful handling to ensure

comparability. `LanguageTool` is case-sensitive, meaning that inputs such as “Fatima” and “fatima” may produce different outputs, whereas `Hunspell` and `SymSpell` normalise inputs to lowercase prior to lookup. Names were therefore passed to `LanguageTool` using natural capitalisation to reflect typical user input.

A further distinction arises in the handling of non-Latin scripts. Initial experiments were conducted using a fixed English configuration (`en-US`); however, this was later replaced with automatic language detection (`language=auto`) to better utilise the full capabilities of `LanguageTool`. In this configuration, the system attempts to detect the language of each token and applies the most appropriate of its language dictionaries where possible. When no suitable language can be confidently identified, the system abstains from providing corrections, resulting in certain inputs being passed through without modification. This behaviour does not indicate recognition, but rather selective non-intervention, and can lead to inflated "recognition" rates for non-Latin names in Condition A. Consequently, for non-Latin script inputs, only the Latin condition provides a valid comparison to the other tools.

3.3.3 Recognition and Correction Suggestions

To ensure consistent large-scale processing, an automated pipeline was developed to apply each spell-checking tool to the enriched dataset under the dual-condition framework. The pipeline operates in sequential stages, beginning with dataset construction and preprocessing, followed by enrichment, and then independent parallel execution of each spell-checking system. Each tool processes the full dataset separately, producing its own set of recognition and correction outputs before results are combined in a final merge step. Parallel execution ensures efficient reproducible processing of the full dataset and allows for opportunity to easily extend the project to include further tools. No evaluation or interpretation is performed within this pipeline; rather, it serves to systematically generate the data used in subsequent analysis.

For each tool, names were evaluated under both conditions (original and transliterated), producing binary indicators of recognition (`*_orig_known`, `*_latin_known`) and corresponding correction suggestions (`*_orig_correction`, `*_latin_correction`) for names flagged as incorrect.

To analyse patterns in correction behaviour, each suggested correction was immediately cross-referenced against the full enriched dataset using a lookup constructed during processing. This produced additional boolean indicators (`*_correction_match`) flagging whether the suggested correction corresponds to another name present in the dataset, along with its associated origin metadata ("top_country", "name_script", "ethnicolr_race", "langdetect_lang").

Because it is infeasible to construct an exhaustive list of valid personal names, this measure serves as a proxy for identifying potential patterns of “correction nudging,” as unrecognised names may be systematically replaced with alternatives that are more frequent or better represented within the underlying lexicon. In particular, if a name originating from a Global Majority origin is corrected to a form that also appears in the dataset, this may indicate a shift toward more commonly attested or standardised name

forms, which are disproportionately drawn from European-dominant corpora.

3.4 Evaluation Methodology and Parameter Choices

The evaluation framework used in this study involves a series of design decisions intended to balance statistical robustness, cross-cultural comparability, and the risk of excluding low-representation groups. These choices are not arbitrary, but reflect trade-offs between precision, coverage, and the uneven distributional structure of the underlying data. The following section outlines how categories are defined, how observations are included in the analysis, and how statistical uncertainty is handled in the interpretation of results.

Top-country assignment threshold. Country attribution is based on a probabilistic top_country assignment, with a threshold of $p \geq 0.7$ used to define a strong geographic association. This reflects a standard thresholding approach in probabilistic classification, balancing assignment confidence with coverage under label uncertainty. The goal is to retain globally distributed naming patterns while avoiding excessive ambiguity in categorical assignment.

Geopolitical standardisation. Following country assignment, all geographic labels are mapped to regions using the United Nations M49 classification system[31]. This ensures consistency in how countries are grouped and avoids ambiguous regional definitions that may introduce inconsistency or implicit researcher-defined bias. It also enables results to be situated within a globally recognised statistical framework defined by the United Nations Statistics Division.

Sample thresholds. To ensure that comparisons remain statistically meaningful while retaining broad geographic coverage, a minimum threshold of $n \geq 100$ was applied for inclusion in country-level visualisations. This step determines which groups are represented in the final plots and is therefore central to the structure of the analysis. At this sample size, 95% Wilson confidence intervals for proportions typically remain within approximately ±10 percentage points under worst-case variance conditions, providing sufficient precision for comparative interpretation. At the same time, this threshold prevents instability in very low-frequency groups, while still preserving near-complete geographic coverage. In practice, only one country group falls below this threshold (El Salvador, $n = 70$) and is therefore excluded from the corresponding visualisations. This reflects a broader concern in fairness-oriented data analysis, where low-frequency groups are at risk of being omitted through aggregation choices, even when their exclusion appears numerically minimal [32].

Confidence interval estimation. All proportion estimates are reported using 95% Wilson confidence intervals. The choice of Wilson intervals reflects the structure of the data, which is highly imbalanced and includes many moderate/small sample strata when aggregated geographically. Compared to standard normal approximation intervals, the Wilson method provides more reliable coverage across the full range of proportions. This makes it more appropriate for sparse and highly imbalanced categorical distributions such as those arising in cross-cultural name data [33]. These intervals provide the uncertainty bounds visualised throughout the results.

Overall, the use of interval-based reporting rather than point estimates reflects the study's aims to not simply rank groups, but to compare distributions under explicit uncertainty, ensuring that observed differences are not overstated in sparse data regimes.

Multivariate analysis and confounding variables. To ensure that observed disparities in recognition rates are not merely artefacts of the name structure or the classification process, the analysis incorporates controls for potential confounding variables. Logistic regression is employed to evaluate the impact of region and script while holding constant factors such as name length (character count) and the confidence score of the probabilistic origin assignment (`top_country_prob`). This allows for a more rigorous assessment of whether bias is systemic across cultural groups or driven by specific technical properties of the input strings.

Effect size reporting. Effect sizes are reported using Cohen's h, which quantifies differences between proportions independently of sample size. Cohen's h is used because it provides a sample-size invariant measure of the difference between proportions, allowing disparities to be interpreted in terms of practical magnitude rather than detectability alone. This is particularly important in fairness-oriented evaluation, where the central question is not whether differences exist (which large datasets will almost always confirm), but whether those differences are meaningfully large in behavioural or systemic terms.

Multiple hypothesis correction. Where statistical comparisons are made across multiple groups, the Benjamini-Hochberg procedure is applied with a significance threshold of $p < 0.05$. This controls the false discovery rate rather than the family-wise error rate, making it more appropriate for exploratory analyses involving a large number of correlated comparisons [34]. This ensures that observed significant effects are not inflated by the volume of tests performed.

Overall interpretation. Taken together, these methodological choices define the structure through which the results are constructed: how names are assigned to categories, how those categories are grouped geographically, which observations are included in the analysis, and how uncertainty and effect size are quantified. Each step reflects a trade-off between statistical precision and representational coverage, rather than a purely technical optimisation decision.

3.5 Ethical Considerations

Research involving names and associated demographic attributes carries inherent ethical risks, particularly when names are used as proxies for racial, ethnic, or national origin. Such inferences, if misapplied, could be repurposed to identify or profile individuals. As boyd and Crawford[35] illustrate in the context of large-scale social media analytics, even seemingly neutral computational processes can reinforce structural inequalities and amplify social biases. These risks are particularly dangerous in contexts such as immigration enforcement, racial profiling, or mass surveillance.

It is important to clarify that this study does not create a tool for individual-level demographic prediction. All analyses are conducted at an aggregate, distributional

level: no system in this project takes a name as input to output an origin prediction for operational use. Rather than creating a new tool, pre-existing classification tools such as `ethnicolr`[29] and `langdetect`[30] are employed to group names into categories that allow systematic, comparative evaluation of spell-checking performance. By constraining inference to group-level statistics, the project avoids generating outputs that could be directly used to profile or target individuals.

Moreover, the study's methodological framing explicitly uses origin classification to critique technological systems, not to enhance demographic identification. Observed disparities in recognition or correction across origin groups highlight limitations of existing spell-checkers and underscore the need for more inclusive computational tools, in line with responsible NLP principles [21, 36]. Rather than to generate actionable demographic information about people, metadata is utilised to detect bias across systems.

The ethical challenges of categorisation are also acknowledged. Grouping names by ethnicity or country of origin in order to quantify bias constitutes a form of racial or national categorisation, which itself can be sensitive or problematic. As Blodgett et al. note, bias measurement necessarily involves choices that risk reproducing the very social categories one seeks to critique [21]. This work recognises such limitations: the grouping of names is a methodological convenience to enable quantitative comparison, not an endorsement of rigid social classification. It is crucial to interpret group-level findings with this constraint in mind.

Finally, the study is guided by the principle that research on algorithmic bias should prioritise transparency, accountability, and societal benefit. By making both data processing and analysis procedures explicit, and by framing demographic inference strictly as an analytical tool for critique rather than operational use, this project seeks to contribute to the broader discourse on ethical and responsible evaluation of computational systems, particularly in the context of human name processing[8].

Chapter 4

Results

4.1 Dataset Overview

4.1.1 Composition and Coverage

The final dataset used for analysis consists of 571,193 unique names, extracted from an initial pool of 727,352 following the application of cleaning procedures (see Section 3.1). These procedures removed entries containing digits, extraneous punctuation, and non-linguistic artefacts, while preserving common naming conventions such as hyphens and apostrophes. The reduction in the dataset reflects the exclusion of entries that would be uniformly flagged by spell-checking systems irrespective of linguistic origin, and therefore would not contribute meaningful variation to the analysis.

The dataset exhibits substantial linguistic diversity, spanning multiple scripts and writing systems. The majority of names are represented in the Latin script, accounting for 81.2% of the dataset. The remaining names are distributed across a range of non-Latin scripts, with the largest groups including Arabic (5.65%), Cyrillic (4.66%), CJK (3.55%), and Hangul (2.16%). This distribution establishes an important baseline for subsequent analysis, as it reflects both the global reach of the underlying Facebook data source and the imbalance in representation across scripts. In particular, the predominance of Latin-script names mirrors the structure of many real-world language technologies, which are often developed and evaluated primarily on Latin-based corpora.

Each name is associated with a probabilistic country-of-origin distribution. The most likely country (`top_country`) is used as a proxy for geographic origin throughout the analysis. The distribution of `top_country` values is uneven, with higher concentrations in countries with large Facebook user bases, such as South Africa and Brazil. As a result, these frequencies reflect platform-specific demographics rather than global population distributions and should be interpreted accordingly.

To assess the reliability of these assignments, the distribution of `top_country_prob` values was examined. A total of 250,976 names (43.9%) exhibit a strong assignment, defined as `top_country_prob` > 0.7, indicating a clear dominant association. The remaining names reflect more ambiguous or globally distributed naming patterns, which

is expected given the cross-cultural reuse and migration of personal names. As an additional validation signal, langdetect's language detection was compared against the official languages of each name's most probable country. Agreement was observed for 98,430 names (17.2%), providing a conservative lower-bound indicator of alignment between inferred country and linguistic context. The relatively low agreement between these tools likely reflects the limitations of using `langdetect` on such short inputs. As such, this measure is interpreted as a proxy for annotation consistency rather than ground truth.

Finally, coverage differs substantially between Latin and non-Latin scripts. While Latin-script names are directly processable by all evaluated tools, non-Latin names require transliteration to enable comparison within a shared representation. This distinction underpins the dual-condition experimental design and is central to separating script-level effects from lexical bias in the following sections.

4.1.2 Enrichment Quality and Limitations

The dataset enrichment process introduced annotations for inferred origin (`ethnicolr_*`), language (`langdetect_*`), and country-level linguistic context (`top_country_langs`). While these features enable multi-dimensional analysis, each carries limitations that affect how results should be interpreted.

The `ethnicolr` classifier assigns each name to one of a fixed set of broad ethnic or regional categories[29]. For Latin-script names, classification is performed directly on the original input, while non-Latin names are classified on their transliterated form (`name_latin`). Coverage is effectively complete, with only 19 names failing to receive any classification. However, the resulting distribution reflects the constrained label space of the model rather than the full diversity of the dataset.

This is most clearly illustrated by the overrepresentation of the *British* category. While only 5,745 names (1.01%) are associated with the United Kingdom or Ireland based on `top_country`, a total of 97,768 names (17.1%) are classified as *British* by `ethnicolr`. This discrepancy arises because the classifier encodes broad ethnic groupings rather than country-level distinctions, and does not distinguish between many closely related European identities (e.g. Irish vs British). Importantly, this is not simply a limitation but a consequence of the model's design: `ethnicolr` operates at a coarser level of abstraction, providing a mid-level grouping that is more informative than script alone, but at a lower granularity than country or language. As such, its outputs are best interpreted as proxies for perceived ethnic similarity rather than precise geographic origin.

Language identification using `langdetect` provides a complementary but less stable signal[30]. Due to the short length of name tokens, predictions are inherently noisy, particularly at lower confidence levels. To mitigate this, each name was repeated five times prior to detection, but some instability remains. To assess this behaviour, outputs were stratified by confidence (`langdetect_prob` ≥ 0.7 vs. < 0.7). While the overall distribution of predicted languages remains broadly similar across both groups, certain languages become disproportionately represented in low-confidence

cases. In particular, English exhibits a bias ratio of 1.27, indicating that it is substantially more likely to be assigned when the model is uncertain. Similar, though weaker, effects are observed for languages such as Swahili, German, and Indonesian (see Table A.1). These patterns suggest that, rather than defaulting to a single fallback language, `langdetect` distributes uncertainty across a small set of high-frequency or globally prevalent languages. Therefore, its outputs should be interpreted cautiously, as predictions may reflect model uncertainty rather than true linguistic signals, particularly for short inputs such as personal names.

To provide a coarse validation measure, detected languages were compared against the official languages of each name's most probable country. Agreement was observed in 17.2% of cases. While this figure appears low, it is consistent with expectations given the short length of inputs, the multilingual nature of many countries, and the independence of the two signals. As such, this measure is interpreted as a conservative lower-bound indicator of alignment rather than a validator of the ground truth.

Finally, the distribution of `top_country_langs` reflects structural biases in the underlying dataset and its frequency-based assignment process. This effect is illustrated in Table 4.1, which isolates names associated with typically English-speaking countries. Despite sharing a common language, these names are disproportionately attributed to a single dominant country. In particular, the United States accounts for the majority of assignments within this group, while other English-speaking countries appear at substantially lower rates, and some (e.g. Australia and New Zealand) do not appear among the most frequent assignments at all.

Table 4.1: Prevalence of English-Speaking Countries (Anglo-Socio-Group) within the Full Cleaned Dataset

Country	**Count (n)**	**Share of Full Dataset (%)***
United States	25,106	4.40
United Kingdom	4,063	0.71
Ireland	1,682	0.29
Canada	1,331	0.23
Total Anglo-Socio-Group	**32,182**	**5.63%**

* Percentage values represent the proportion of the total cleaned dataset ($N = 571,539$) assigned to a given country as the primary `top_country` prediction.

This pattern does not reflect the absence of English-language names from these regions, but rather the way in which the dataset aggregates frequency signals. Names that are widely used across multiple English-speaking countries are more likely to be attributed to the country with the largest representation in the source data. This results in a concentration of assignments in the United States, even when names are not uniquely associated with that region. A similar concentration effect is observed more broadly across the dataset (see Appendix A.2).

As a result, country labels should be interpreted as reflecting dominant frequency associations rather than precise geographic origin. This limitation is particularly important

for downstream analysis: patterns observed at the country level may partially reflect dataset concentration effects rather than true linguistic or cultural distinctions. For this reason, subsequent analysis prioritises broader groupings (e.g. script, language, and `ethnicolr` categories), where such aggregation effects are less pronounced.

Taken together, these enrichment features provide useful but imperfect proxies for script, origin, and language. The analysis therefore prioritises patterns that are consistent across multiple independent signals, reducing reliance on any single annotation and strengthening the robustness of the findings.

4.2 Recognition Disparities

4.2.1 Overall Tool Performance

Across all tools and conditions, the vast majority of names in the dataset are flagged as misspelled. This is expected, as personal names are typically absent from standard lexicons, and establishes a baseline against which disparities in recognition can be evaluated.

Under Condition B (*Latin transliteration*), recognition rates for the cleaned dataset ($N = 571{,}193$) are 4.90% for Hunspell (27,995 names), 4.16% for SymSpell (23,735 names), and 13.28% for LanguageTool (75,871 names), as shown in Table 4.2. Although proportions are low, these correspond to tens of thousands of recognised names in absolute terms. While absolute recognition remains low across all tools, LanguageTool passes through 2.7-3.2 times more names than either Hunspell or SymSpell. For Hunspell and Symspell, transliteration yields relative improvements on recognition rates of 49% and 58%, respectively, highlighting the extent to which recognition is constrained by script compatibility.

Table 4.2: Overall Recognition Rates by Tool and Condition (Cleaned Dataset, $N = 571,193$)

	Condition A		**Condition B**		**Ratio**
Tool	**Known** (n)	**Rate (%)**	**Known** (n)	**Rate (%)**	(B/A)
Hunspell	18,830	3.30	27,995	4.90	1.49
SymSpell	14,999	2.63	23,735	4.16	1.58
LanguageTool	87,032	15.24[†]	75,871	13.28	0.87

[†] The Condition A rate for LanguageTool is inflated by selective abstention on non-Latin-script inputs and is not directly comparable to other tools.

Even in the most permissive configuration (LanguageTool, Condition A), only approximately 1 in 6-7 names are permitted, meaning that just under 85% of names are still treated as errors.

The relationship between Condition A (*original script*) and Condition B (*Latin transliteration*) differs substantially across tools. For Hunspell and SymSpell, recognition

improves under transliteration, reflecting their reliance on ASCII-based dictionaries. Names written in non-Latin scripts are largely unrecognisable in their original form, but transliteration brings them into a space where partial matching becomes possible. Alternatively, LanguageTool exhibits the opposite pattern, with a higher apparent recognition rate under Condition A. This effect, however, is not indicative of improved performance. When operating in automatic language detection mode (`language=auto`), LanguageTool frequently abstains from flagging tokens in scripts it cannot confidently classify. For several non-Latin scripts, this results in near-complete pass-through behaviour, inflating the apparent recognition rate. As discussed in the Methodology, this should be interpreted as abstention rather than successful recognition. Consequently, Condition B provides the only meaningful basis for cross-tool comparison.

Within Condition B, the higher recognition rate of LanguageTool reflects its multilingual design. Unlike Hunspell and SymSpell, which rely primarily on English-centric lexicons, LanguageTool can apply multiple language models depending on the detected input. This broader coverage allows it to recognise a wider subset of transliterated names, though performance remains limited overall.

Hunspell and SymSpell exhibit similar behaviour, with relatively low recognition rates and a consistent gap between them. SymSpell performs slightly worse, likely due to its reliance on frequency-based string matching without morphological expansion, whereas Hunspell's affix rules can occasionally recover valid forms.

Despite these differences, the overall pattern is consistent: all tools fail to recognise the majority of names. The key question is therefore not whether recognition fails, but whether this failure is evenly distributed. The following sections examine how recognition varies across scripts and origin groups, where systematic and statistically significant disparities emerge.

4.2.2 Statistical Significance and Effect Sizes

To assess whether the observed differences in recognition are meaningful, a combination of statistical tests and effect size measures was used. Chi-squared tests assess whether variation across scripts and origin groups is unlikely to arise by chance, while effect size metrics (Cohen's h) quantify the magnitude and direction of these differences. Logistic regression is then used to examine whether these patterns persist when controlling for factors such as name length and geographic concentration.

Table 4.3: Chi-Squared Independence Tests for Recognition Rate Variation

Tool & Condition	**Grouping Variable**	df	χ^2	p
Hunspell B	Script family	9	3,462.8	< 0.001
SymSpell B	Script family	9	2,821.2	< 0.001
LT-auto B	Script family	9	83,971.9	< 0.001
Hunspell B	Origin category (Latin)	12	4,944.9	< 0.001
SymSpell B	Origin category (Latin)	12	3,687.2	< 0.001
LT-auto B	Origin category (Latin)	12	8,294.8	< 0.001

Across all tools, chi-squared tests indicate that recognition rates vary systematically across both script families and origin categories (all $p < 0.001$; seen in Table 4.3). While this confirms that the differences are non-random, the large sample size means that even relatively small deviations are statistically significant. As such, significance alone is not sufficient, and effect sizes provide a more meaningful basis for comparison.

We therefore compute Cohen's h for each script, relative to Latin-script names, to assess the magnitude of disparities (Figure 4.1; Table 4.4). This reveals clear and consistent patterns across tools. Hunspell and SymSpell show medium-magnitude under-recognition for Arabic, Hebrew, Hangul, and CJK scripts, even after transliteration. As shown in Figure 4.1, the contrast against LanguageTool is particularly pronounced for CJK and Hangul, where it diverges sharply from the dictionary-based tools.

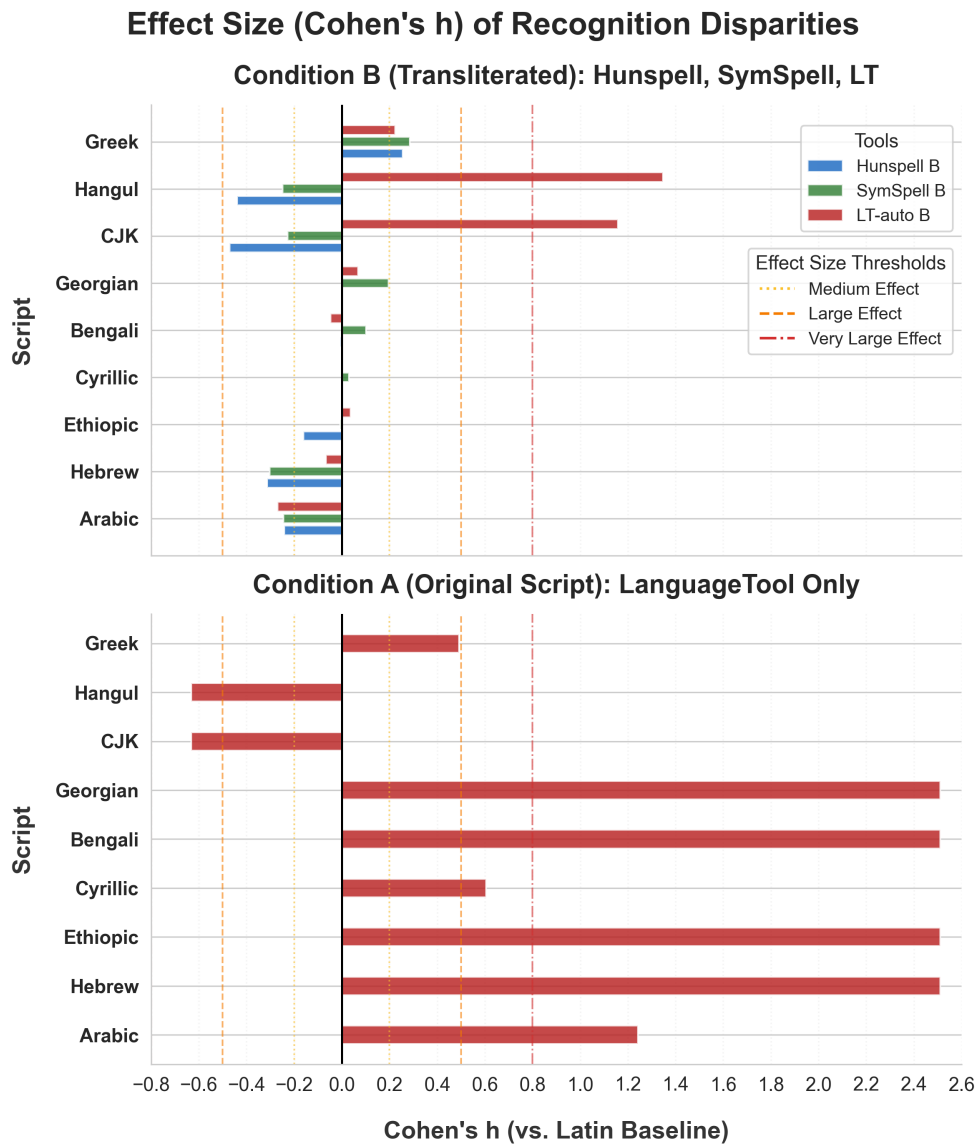

Figure 4.1: Cohen's h effect sizes by script for Condition B (transliterated) and LanguageTool's Condition A (original), relative to Latin-script recognition rates across tools.

This divergence highlights a fundamental difference in behaviour. Dictionary-based

Table 4.4: Cohen's h Effect Sizes by Script Family in Condition B (transliterated)

Script Family	Hunspell B	SymSpell B	LT-auto B
CJK	-0.47	-0.23	1.16
Hangul	-0.44	-0.25	1.35
Hebrew	-0.31	-0.30	-0.07
Arabic	-0.24	-0.25	-0.27
Ethiopic	-0.16	-0.01	0.04
Bengali	-0.01	0.10	-0.05
Cyrillic	0.00	0.03	0.00
Greek	0.25	0.29	0.22

Negative values indicate lower recognition than the Latin origin script baseline.

tools remain constrained by Latin-centric lexical coverage, whereas LanguageTool's multilingual architecture interacts with transliterated forms in a way that can inflate recognition for certain scripts. Arabic is the only script with consistently negative effects across all tools, while Greek shows consistent positive effects, suggesting that some transliteration patterns align more closely with existing lexicons than others.

To assess whether the observed disparities are driven by structural characteristics of the data, logistic regression models were fitted with script as the primary explanatory variable. Latin-script names are treated as the reference category, allowing all other script coefficients to be interpreted as differences in recognition likelihood relative to this baseline.

Two additional variables are included to account for general properties of names: name length, and the probability associated with the most likely country of origin. These capture behaviour that may influence recognition, such as longer names having more spelling variation and thus being less likely to match dictionary entries, or names with strong geographic concentration being more/less aligned with English-focused lexicons.

Figure 4.2 presents the resulting odds ratios on a logarithmic scale, adjusted for name length and geographic concentration. Across all functional models, increased name length and higher geographic concentration are consistently associated with lower odds of recognition ($p < 0.001$), suggesting that tools struggle with both structural complexity and names highly specific to non-Latin locales. Even when accounting for these factors, profound script-based disparities persist. Arabic and Hebrew are associated with significantly lower odds of recognition across nearly all tools, while Greek consistently exhibits a performance bonus relative to the Latin baseline. Notably, recognition for East Asian scripts (CJK and Hangul) remains highly polarised: while effectively invisible to Hunspell and SymSpell, they show a significant positive association with recognition in the LT-auto B model.

This indicates that the disparities observed are not simply a by-product of general name characteristics, but are specifically linked to script-related differences in how names are processed by each tool.

Taken together, these analyses show that recognition disparities are not only statistically

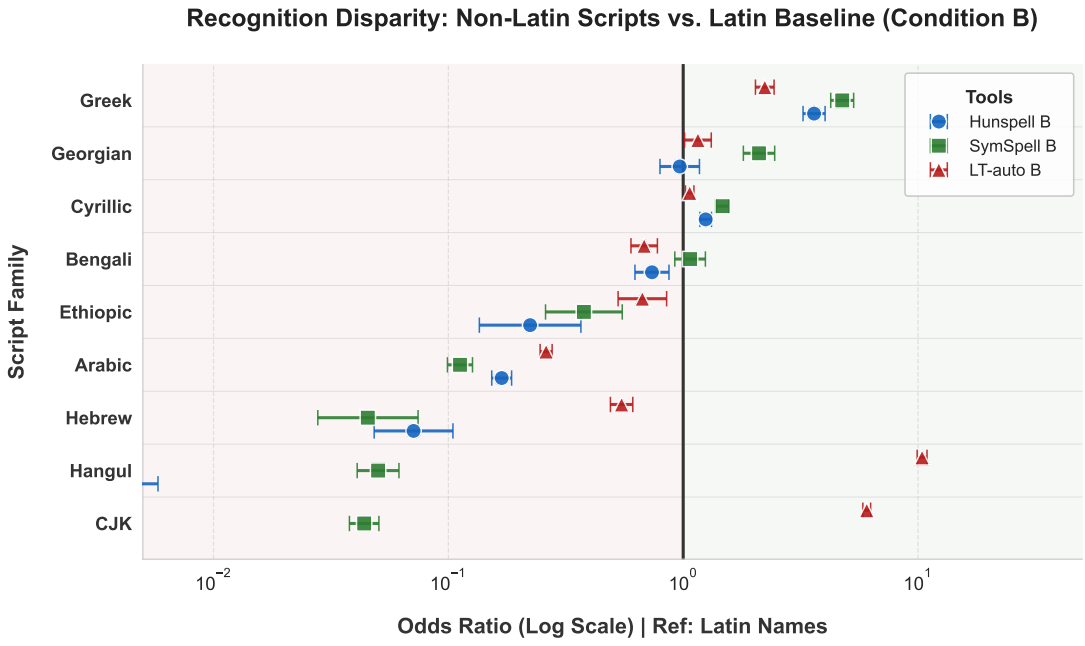

Figure 4.2: Logistic Regression: Recognition Likelihood by Script under Condition B (transliteration).
Note: A small vertical jitter has been applied to the markers within each script category to ensure that the confidence intervals for all tools remain clearly visible and do not overlap.

robust, but also of meaningful magnitude and resilient to common structural factors. The following sections examine these patterns in more detail, focusing on how they manifest across specific scripts and origin groups.

4.2.3 Script-Level Effects and the Transliteration Advantage

The analysis of overall recognition rates indicates that script is a primary factor in determining whether a name is identified as valid by spell-checking systems. However, the magnitude and direction of this effect varies substantially across tools and is largely shaped by whether inputs are evaluated in their original script (Condition A) or in transliterated Latin form (Condition B).

Across all systems, recognition rates differ materially between the two conditions, with the most consistent improvements observed, after transliteration, in lexicon-constrained systems. For Hunspell and Symspell, successful recognition is fundamentally contingent on token overlap with entries encoded in a Latin-script dictionary. As a result, non-Latin scripts in Condition A lie entirely outside the recognition space of these systems. Transliteration successfully acts as a structural bridge, projecting names into a character space compatible with their underlying lexicons and enabling partial recovery through approximate matching and dictionary lookup. This effect is reflected in Figure 4.3, which shows a consistent upward shift in recognition rates for both Hunspell and SymSpell across nearly all non-Latin script families under Condition B.

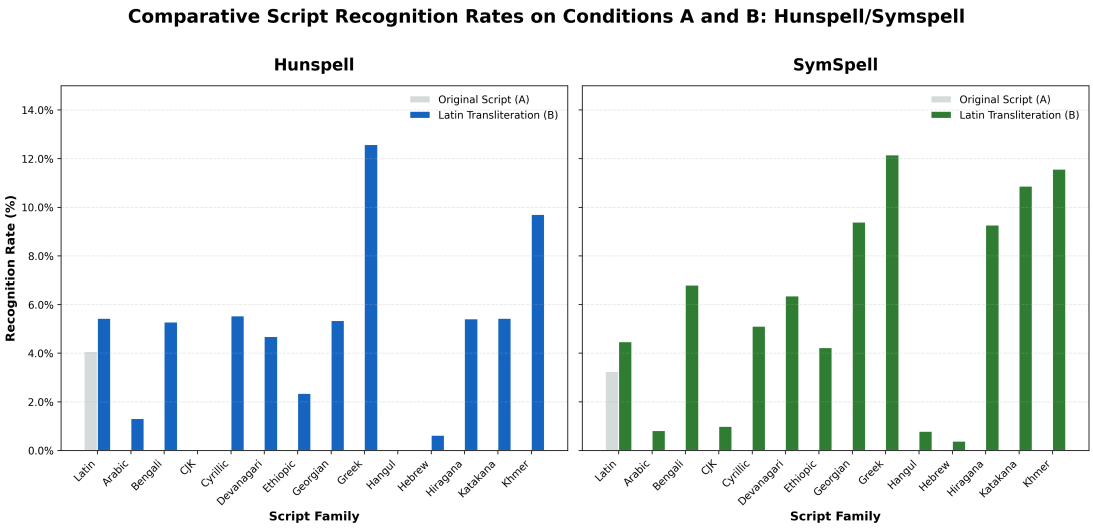

Figure 4.3: Recognition rates for lexicon-constrained tools (Hunspell and Symspell) across Condition A (Original Script) and Condition B (Latin Transliteration), stratified by script family.

In contrast, LanguageTool operates as a hybrid system that incorporates multiple mechanisms, including dictionary resources, rule-based grammar models, and automatic language identification. This introduces an additional decision layer: rather than uniformly attempting to evaluate all inputs, the system may selectively abstain when it cannot confidently assign a language model to a given token. This behaviour is particularly pronounced for non-Latin scripts under Condition A, where low language-identification confidence frequently results in tokens being passed through without meaningful evaluation. As a consequence, the apparently higher recognition rate observed for LanguageTool in Condition A should not be interpreted as improved lexical coverage, but rather as systematic abstention from error flagging in cases where the system cannot reliably classify the input. Condition B therefore provides a more comparable basis for cross-tool evaluation.

To quantify the effect of transliteration more directly, the difference between Condition A and Condition B can be interpreted as a measure of script-constrained recoverability: the extent to which names become accessible to a system once encoded in a Latin-compatible form. Under this framing, Hunspell and SymSpell show a clear transliteration advantage, with increased recognition rates reflecting the partial restoration of dictionary matching once script barriers are removed. However, even after transliteration, absolute recognition rates remain low across both systems, indicating that script compatibility is a necessary but not sufficient condition for recognition.

Importantly, the impact of transliteration is not uniform between scripts. East Asian scripts, particularly CJK, exhibit the largest gains under Condition B, in comparison to their near-zero recognition in Condition A, reflecting successful transliteration of these ideographic scripts. Arabic and Hebrew show more modest improvements, suggesting that their transliterated forms remain less aligned with the lexical patterns present in Latin-script dictionaries. Cyrillic occupies an intermediate position, reflecting partial

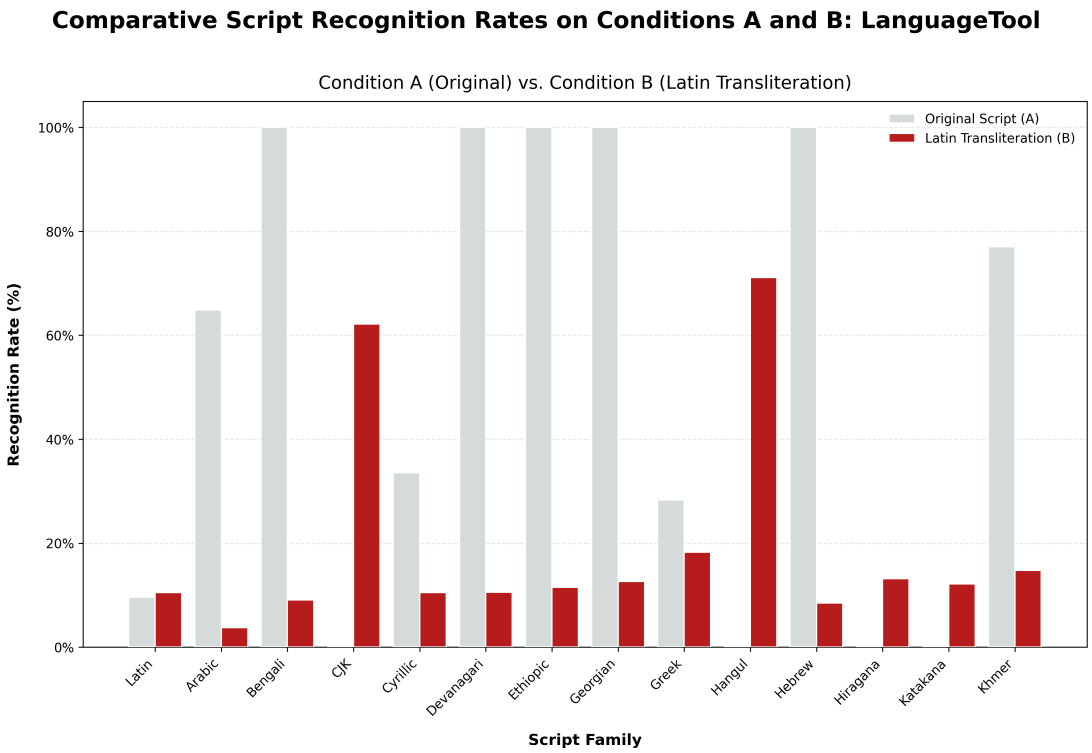

Figure 4.4: Recognition rates for hybrid-tool LanguageTool across Condition A (Original Script) and Condition B (Latin Transliteration), stratified by script family.

overlap between transliterated forms and existing Latin-script.

A notable and unexpected pattern emerges for Greek script, which consistently exhibits positive deviations relative to the Latin-script baseline across all tools in Condition B, as seen in Figure 4.1. In other words, transliterated Greek names are not merely recovered at higher rates than their original-script counterparts, but are often recognised more successfully than names that are already Latin-script. This suggests that Greek-origin names, when transliterated into Latin form, may align particularly well with existing dictionary entries and culturally embedded name inventories. One possible explanation is that Greek naming conventions, especially those represented in widely used corpora and historical datasets, overlap strongly with traditional Latin naming structures that are already well-represented in spell-check lexicons. Whereas, the Latin-script category itself is highly heterogeneous, inclusive of a wide range of linguistic and cultural naming systems, which increases variability and reducing average lexical alignment.

This observation introduces an important refinement to the interpretation of script effects: Latin script does not represent a linguistically uniform or privileged category, but rather a high-variance space containing both highly recognisable and systematically under-recognised naming traditions. Greek, in this context, may function as a comparatively "low-variance" subset whose transliterated forms map more consistently to established lexicon structures.

Together, these patterns highlight a key distinction between two sources of constraint. Script-level effects determine whether a name is structurally accessible to a system at all, while lexical coverage determines whether it is recognised once that barrier is removed. Transliteration isolates the former by standardising script representation, but

does not eliminate the latter, which continues to shape recognition outcomes within the shared Latin space. This separation motivates the following section, which focuses on variation in recognition rates within Latin-script names only. In this restricted setting, script-level effects are held constant, allowing lexical and cultural biases embedded in dictionary composition to become the primary shaping factor of observed disparities.

4.2.4 Lexical Bias within Latin Script

Having isolated script-level effects in the previous section, this analysis now focuses exclusively on Latin-script names under Condition B. By restricting the input space in this way, differences in recognition can be attributed primarily to lexical coverage rather than structural incompatibility between writing systems.

Recognition rates vary substantially across `ethnicolr`'s origin categories, even within a shared Latin encoding. Figure 4.5 shows these rates with 95% Wilson confidence intervals, for all three tools, revealing clear and consistent disparities across both lexicon-constrained systems.

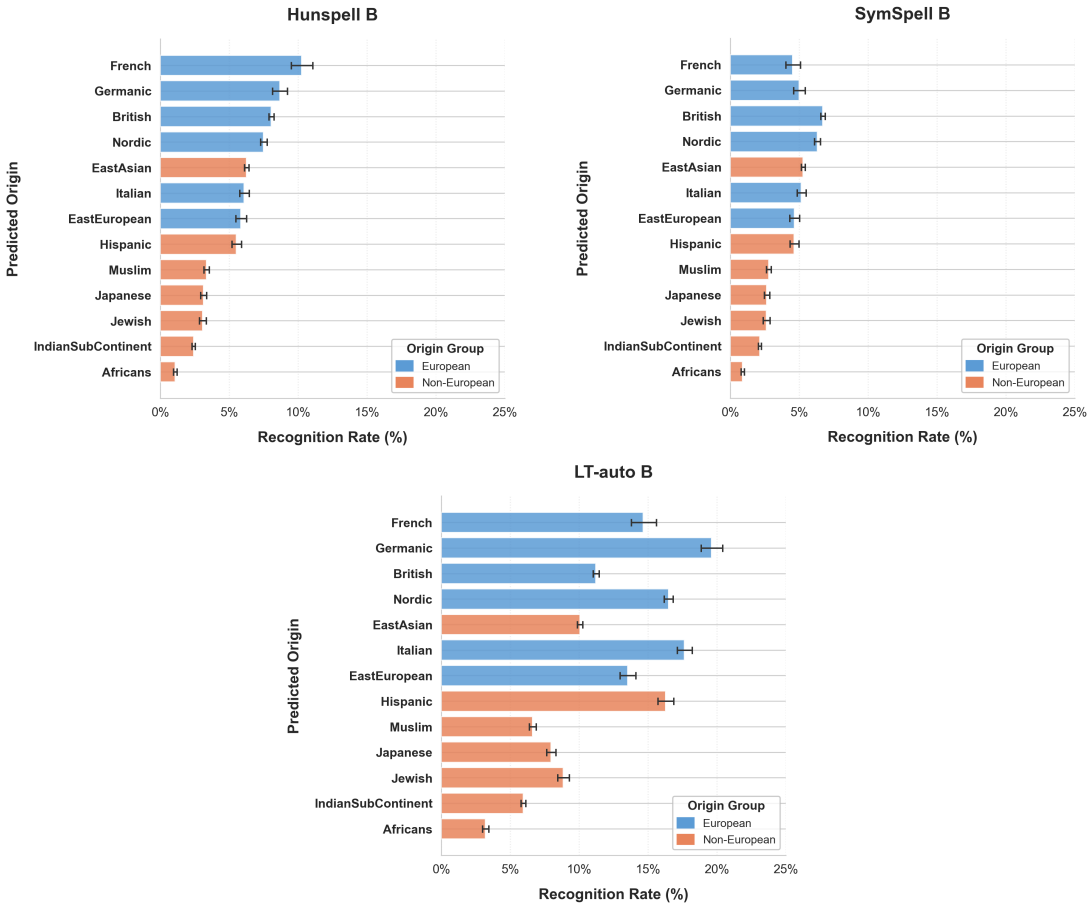

Figure 4.5: Comparison of recognition rates across tools for Latin-script names (Condition B), stratified by predicted ethnicolr origin region. Coral bars indicate European-centralised origin groups; blue bars indicate Global Majority widely distributed origin groups.

A clear pattern emerges in Figure 4.5, particularly for the lexicon-constrained sys-

tems. For both Hunspell and Symspell, European origin categories cluster at higher recognition rates, while globally distributed categories are concentrated at the lower end.

Although LanguageTool shows a wider range, a similar structure is evident. European-origin categories remain amongst the highest-performing groups, though boundary is less distinct, with categories such as Hispanic and East Asian achieving comparable or higher recognition. This suggests that LanguageTool's multilingual design has successful recognition and analysis of select non-European languages.

Despite the variation in the upper half, the lowest-performing categories are consistent across all three tools. African, Indian Subcontinent, and Jewish groups remain well below all European-origin group, indicating a persistent and non-tool specific pattern of under-representation within the underlying corpora.

The confidence intervals shown in Figure 4.5 support the stability of these patterns. At the extremes, intervals are clearly separated, indicating that the observed disparities are robust rather than driven by sampling variation. More broadly, narrow intervals for large groups lend precision to these comparisons, reinforcing that the differences reflect underlying structural patterns in recognition behaviour.

Some overlap in confidence intervals is observed within both lower- and upper-performing clusters, indicating that precise ordering between nearby groups is less certain. However, this does not undermine the central finding as the separation between the higher-performing European-origin groups and lower-performing globally distributed groups remains stable across all three tools, even when accounting for uncertainty.

To formally assess the direction and statistical significance of these differences, Figure 4.6 presents pairwise comparisons between each origin category and the British reference group under Condition B. Differences are expressed in percentage points and adjusted using the Benjamini-Hochberg procedure.

Across all three tools, most globally distributed origin categories exhibit significantly lower recognition rates than the British reference group, with the largest gaps observed for African and Indian Subcontinent categories.

In contrast, European-origin categories show smaller and less consistent deviations. Some groups remain close to the British baseline, while others exceed it depending on the tool. This reinforces that the disparities observed in Figure 4.5 are statistically robust as well as descriptive.

To further contextualise these patterns, Figure 4.7 aggregates recognition rates at the UN M49 regional level, showing that the variation observed at the origin-category level extends to a broader geographic structure.

At the regional level, a clear hierarchy is observed across all three tools. Northern America shows the highest and most stable recognition rates, while Europe follows with substantially greater spread. This may reflect that the Northern America region including only Canada and United States which are majority English-speaking countries whereas the European region group is highly heterogeneous.

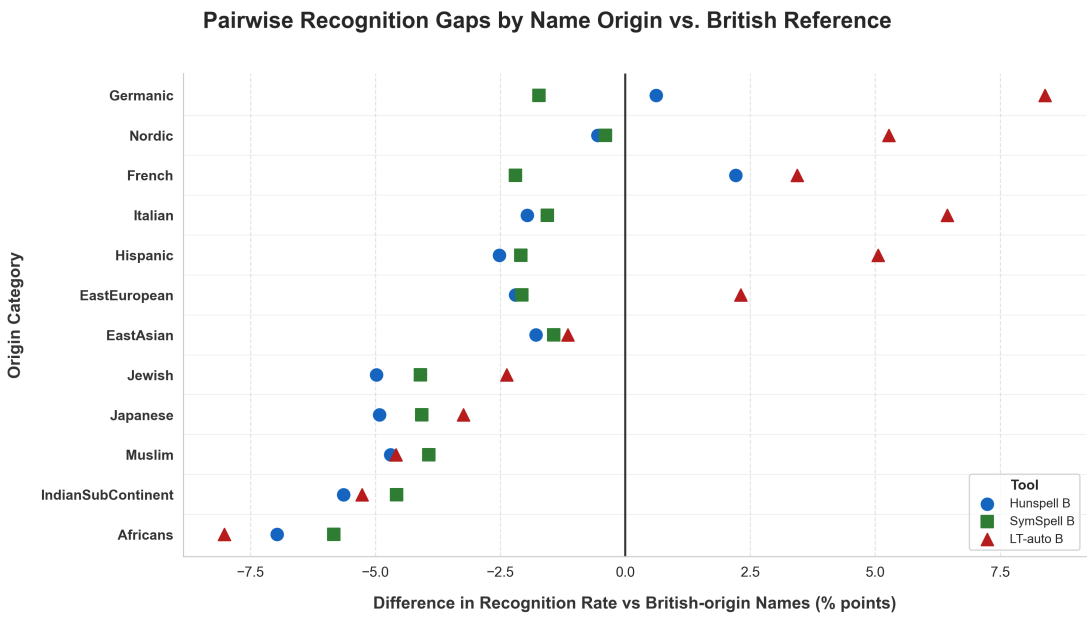

Figure 4.6: Difference in percentage recognition rate of each ethnicolr origin category vs British-origin category (Latin-script only, Condition B). Negative values indicate lower recognition than British.

Recognition declines progressively across remaining regions, with several falling below the lower bound observed in Europe. While LanguageTool generally achieves higher overall recognition, the relative ordering of the regions and their spread remains similar across all tools, indicating similar behaviour across lexical conditions. Overall, the regional aggregation mirrors the origin-category results showing that recognition within Latin script follows a consistent and structured geographic gradient.

To explore whether these disparities may be partly driven by alignment between a name's linguistic form and its associated country, we compared recognition rates for names where the predicted language matches an official language of the top-country prediction against those where it does not. Across all three tools, names whose predicted language aligns with the official language of the associated country are recognised at higher rates, see Figure A.1. This pattern is consistent, though it should be interpreted cautiously, as it may reflect the types of languages that `langdetect` is more able to confidently identify, as well as biases in the underlying dataset distribution, rather than a direct causal effect.

Recognition remains unevenly distributed even among names that align with their country's official language, indicating that language alignment alone cannot account for the full pattern of variation. To isolate the extent to which these disparities reflect lexical bias rather than confounding factors, a logistic regression analysis was conducted using recognition as the outcome variable, with origin category as the primary predictor and controlling for name length and top-country prediction confidence.

Across all three tools, most origin categories retain significantly lower odds of recognition relative to the British reference group after controlling for these factors, indicating that the disparities observed are not solely attributable to differences in name length or

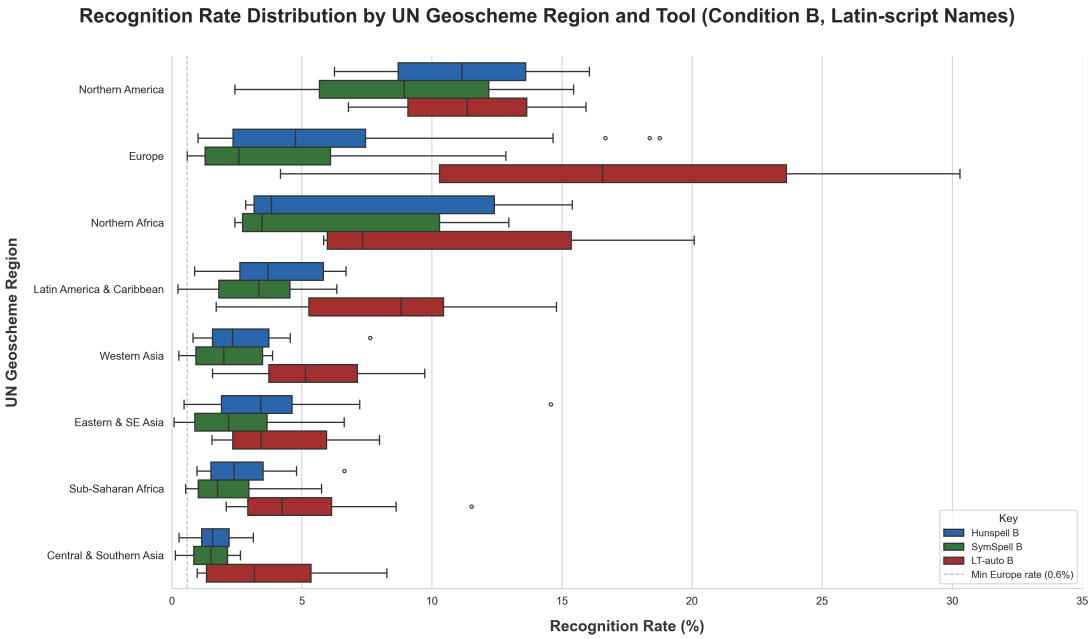

Figure 4.7: Box Plot of recognition rates grouped by UN geoscheme region.

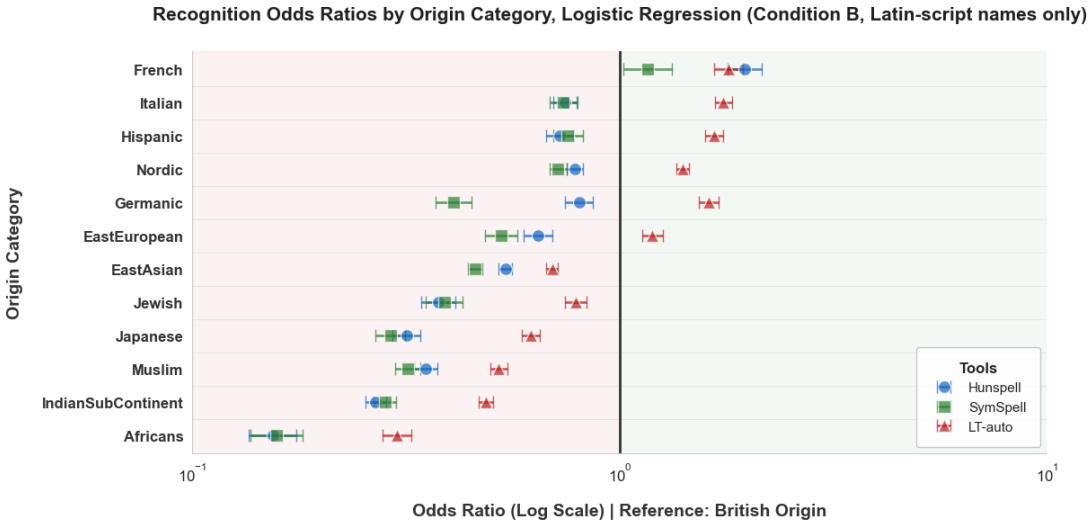

Figure 4.8: Controlled for name length and top-country probability.

prediction confidence, but persist as a function of lexical representation.

One notable exception is the French category, which exhibits higher odds of recognition relative to British across all tools. For LanguageTool, several additional European-origin categories also show elevated odds, reflecting its broader multilingual coverage.

Despite some variation, Global Majority origin categories consistently show reduced recognition likelihood, even under controlled conditions. This provides strong evidence that the disparities observed throughout this section are rooted in systematic differences in lexical coverage rather than artefacts of input structure or model confidence.

4.3 Correction Behaviour

This section explores the patterns of behaviour in names flagged as unrecognised, providing insight into potential patterns of cultural erasure. All primary visuals in this section utilize UN M49 regional groupings assigned via the dataset's `top_country` field. To ensure the robustness of these findings across different classification schemas, corresponding visuals grouped by `ethnicolr` racial categories are provided in Appendix A.3; these figures demonstrate broadly similar patterns of behaviour to those discussed below.

4.3.1 Correction Rates and Dataset Matching

Figure 4.9 presents the proportion of unrecognised names for which each tool provides a correction, aggregated by UN M49 region. This captures the systems' willingness to generate alternative forms under Condition B. Across all tools, correction availability varies substantially, with a consistent ordering in which SymSpell offers suggestions most frequently, followed by LanguageTool and then Hunspell. In contrast to the recognition results, however, this variation does not follow a clear regional hierarchy. Regions with lower recognition rates do not systematically receive more or fewer correction attempts, suggesting that correction behaviour is not strongly conditioned on the same factors that drive recognition performance.

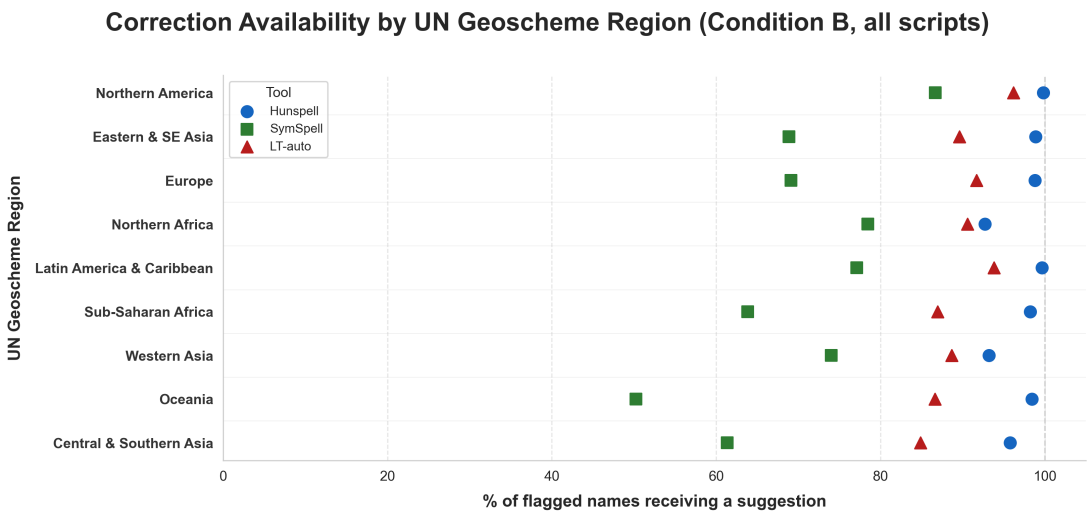

Figure 4.9: Each tool's rate of corrections, grouped by UN M49 region

To assess the quality of these suggestions, Figure 4.10 reports the proportion of corrections that match known names within the dataset, again aggregated by region. This check serves as a proxy for whether generated corrections correspond to plausible name forms rather than arbitrary strings, as it is not possible to create a rule-based criterion for identify whether a string constitutes a personal name.

Across all tools, only a subset of suggestions correspond to known names, with moderate variation by region. While some regions exhibit slightly higher matching rates, the overall pattern remains relatively stable, indicating that differences in correction validity are less pronounced than those observed in recognition. This suggests that, although

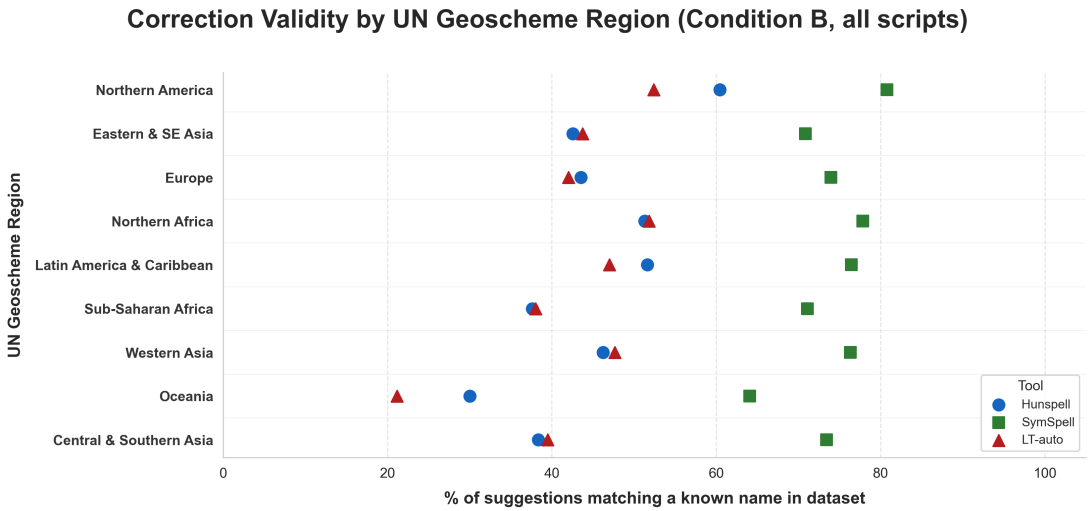

Figure 4.10: Each tool's rate of corrections appearing in dataset, grouped by UN M49 region

systems vary in how often they propose corrections, the likelihood that these correspond to plausible names is more uniform across regions.

This view highlights differences in overall system behaviour. SymSpell produces known-name suggestions at rates consistently above 60% across regions, while Hunspell and LanguageTool show greater variation and lower rates of known-name outputs. Despite these differences, the relative distribution of outcome types remains broadly similar across regions, reinforcing that variation in correction behaviour is driven more by tool design than by geographic grouping.

4.3.2 Directional Bias in Suggestions

Beyond whether corrections are offered, it is also important to consider the direction in which names are transformed. Figure 4.11 examines the distribution of correction targets, showing, for each source region, the proportion of suggestions that map to European-origin names, as well as the proportion that remain within the same origin group.

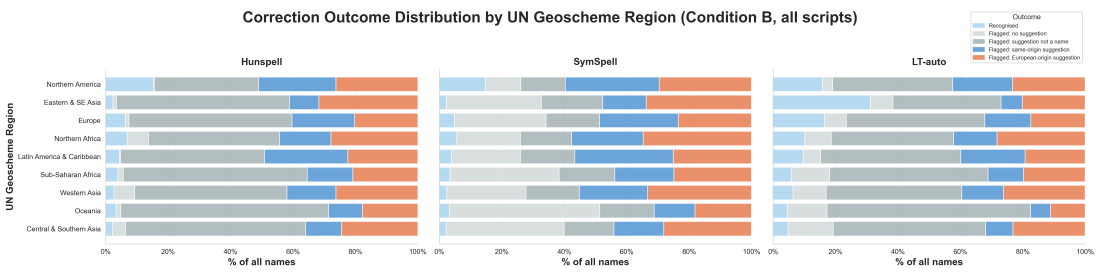

Figure 4.11

A clear asymmetry is visible in the distribution of correction targets across regions. Names originating from European regions are frequently corrected to other European-origin names, with a substantial proportion also remaining within the same origin group.

In contrast, names from many non-European regions are more likely to be mapped to European-origin suggestions, and less likely to retain their original origin in the correction process.

This pattern is not uniform across all groups. Some regions, such as East and South-East Asia, show relatively high rates of same-origin correction, indicating that alternative forms are available within the same lexical space. Others, including Sub-Saharan Africa and parts of South Asia, exhibit lower same-origin retention and a greater tendency for corrections to shift toward European-origin names.

Across all tools, similar directional tendencies are observed, although the strength of the effect varies. The lexicon-constrained systems show closely aligned behaviour, while LanguageTool exhibits greater variation, consistent with its broader multilingual coverage.

To examine these patterns more directly, Figure 4.12 shows a heatmap of region to region name mappings. This shows that European and Northern American names are frequently offered as corrections, across a wide range of source regions.

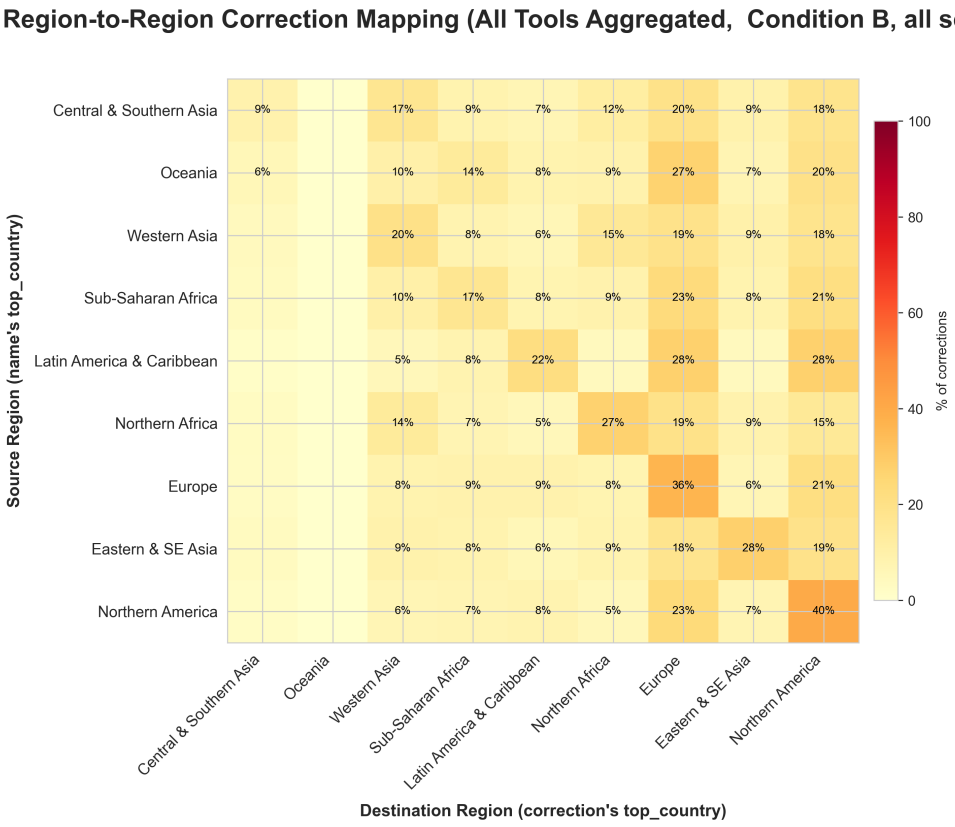

Figure 4.12: Region to region mappings of names to their suggested correction, grouped by UN M49 regions.

Overall, these results indicate that correction behaviour is not neutral with respect to origin. When names are transformed, they are not only normalised toward recognised forms, but are often redirected toward names associated with Global North origins. This extends the recognition findings by showing that disparities persist not only in whether names are recognised, but also in how they are altered when they are not.

Chapter 5

Conclusions

This study examined whether spell-checking systems treat names from different cultural origins differently, and explored the mechanisms through which such differences arise. Using a range of analytical approaches, including script-level, ethnic category and regional aggregation, and controlled regression, recognition rates were found to vary systematically, with disparities persisting under Latin-script conditions and after accounting for key confounding factors. These patterns extend into correction behaviour, indicating that variation is not limited to whether names are recognised, but also how unrecognised names are transformed. Together, the consistency of these findings suggests that the observed disparities reflect underlying structural biases and differences in lexical coverage, rather than isolated system artefacts.

5.1 Completion of Goals

This section returns to the central research questions and summarises how each is addressed by the findings.

RQ1: Detection of disparity. Across all three tools, recognition rates were unevenly distributed across origin groups. Names associated with European and North American origins were consistently recognised at higher rates, while those associated with Global Majority origins and wider global distribution exhibited lower recognition. This pattern was observed across multiple levels of aggregation including script, `ethnicolr` categories, and UN M49 regions. The persistence of this structure indicates that the disparities are systematic, not merely the result of isolated cases or individual system behaviour.

RQ2: Mechanism of bias. By isolating Latin-script names and controlling for factors such as name length and prediction confidence, this study assessed whether disparities could be attributed solely to structural differences between writing systems. The persistence of recognition gaps under these conditions indicates that script alone does not account for the observed variation. Instead, the results point to differences in lexical coverage within a shared Latin representational space, suggesting uneven representation of naming conventions within the underlying lexicons.

RQ3: Correction behaviour. Analysis of correction behaviour shows that disparities extend beyond recognition into the handling of unrecognised names. While tools differ in how frequently they offer suggestions, these differences do not follow the same structured geographic patterns observed in the recognition rates. Conversely, the direction of corrections reveal a more consistent tendency: suggestions are more likely to map names to Global North (European or Northern American) origin names, and less likely to preserve the original naming convention for many Global Majority groups. This indicates that when names are not recognised, they may be systematically shifted toward more frequently represented naming systems.

Taken together, these findings demonstrate that bias in spell-checking systems operates at multiple stages, affecting both whether names are recognised as well as how they are transformed when they are not.

5.2 Limitations of Results

5.2.1 Conceptual and Operational Definitions

Several limitations affect both the interpretation of these results and the strength of the claims that can be made about real-world impact. These arise from the conceptual framing of names as analytical units, the operationalisation of recognition, and the properties of the dataset and tools used.

Most fundamentally, names do not naturally belong to a single country or language. Assigning each name to a single `top_country` or primary language is an operational decision that enables large-scale comparison, but does not reflect how names function in practice. Names frequently migrate across cultures, are adapted across linguistic contexts, and persis in diasporic communities. For example, a name such as "Maria" is widely used across multiple countries simultaneously, and its assignment to a single country reflect the proportional share in the dataset rather than a definitive origin. Similarly, names associated with one region may be widely used elsewhere without this distribution being proportionally captured in the data. The use of a strong `top_country` threshold (probability ≥ 0.7) partially mitigates this by distinguishing more geographically concentrated names from widely distributed ones, but does not resolve the underlying ambiguity. Country and language labels should therefore be interpreted as analytical groupings, rather than claims about cultural ownership or origin.

A related limitation concerns the use of names as proxies for demographic or cultural identity. This approach is consistent with prior bias audits in natural language processing, where name lists have been used to approximate categories such as race or gender [15, 16]. However, this methodology carries well-established limitations. A name does not uniquely determine an individual's identity. As such, the analysis operates at a population or distributional level rather than making claims about individuals. Furthermore, unlike many prior audits which evaluate model outputs against human-labelled ground truth, there is no authoritative standard for whether a name "should" be recognised by a spell-checker. The evaluation is inherently relative, comparing

recognition rates across groups rather than measuring absolute correctness. Although this restricts the interpretation of results, it also aligns with the goal of identifying systematic disparities rather than prescribing normative standards.

Defining what constitutes a "name" presents an additional challenge. Naming conventions vary widely across cultures, and any rule-based definition risks imposing a particular cultural model as universal. Criteria such as minimum length or character restrictions can inadvertantly encode the same assumptions that this study aims to critique. To avoid this, the analysis relies on a large-scale name frequency dataset derived from self-reported names, allowing the definition of naming conventions to emerge from real-world usage rather than imposed rules. This approach improves cultural inclusivity but introduces noise, such as presence of initials, punctuation, and usernames. Cleaning procedures were therefore applied to reduce noise while remaining conservative to avoid the reintroduction of definitional bias.

The operational definition of recognition as "known" measures inclusion in dictionary not name validity. This reflects inclusion within a lexicon rather than any notion of linguistic or cultural validity, nor any tool's awareness of a string as a name. Spell-checkers do not distinguish between personal names and ordinary strings as they lack awareness of this conceptual difference. For example, names such as "Grace" may be recognised because they exist as an English word, while "James" may pass as it is included in the dictionary to historical frequency within English corpora. This creates a structural bias towards names associated with languages that have long-standing representation in these lexicons, particularly English [11]. As a result, the metric captures inclusion within technical systems rather than the legitimacy of a name itself.

5.2.2 Dataset and Tool limitations

As Section 3.1 established the platform-specific nature of the data, the results must be interpreted through the lens of 'digital visibility'. The proportional under-representation of regions like China means the findings regarding those scripts/naming conventions are likely conservative; the true disparity in spell-checking systems may be even more pronounced than this study suggests. While much of this bias is partially mitigated by focusing on relative rates rather than absolute counts, the resulting distributions should still be understood as reflecting a platform-shaped sample rather than a globally representative population.

Second, the dataset does not distinguish between first names, surnames, or full names, as all entries originate from a single user-defined name field. For the single-token subset used in the primary analysis, this ambiguity does not materially affect the evaluation, as spell-checking systems treat all tokens equivalently regardless of name position. Nevertheless, it remains a structural limitation of the dataset, and should be considered when interpreting the broader composition of name types included in the analysis.

Third, the presence of multi-word entries introduces a systematic filtering effect. When passed as single tokens, multi-word names are almost always rejected by dictionary-based systems, regardless of cultural origin, because they do not match lexicon entries. This would artificially inflate error rates for groups whose naming conventions more

frequently include multi-word forms. To control for this, the primary analysis restricts evaluation to single-word names, ensuring comparability across groups.

Finally, it must be acknowledged that the tools used for racial and linguistic classification carry their own inherent biases. Because libraries such as `ethnicolr` rely heavily on administrative records like the Florida voter registration files, the resulting classifications are a reflection of how those specific systems categorize individuals[29]. These labels must be carefully viewed as a proxy that may inadvertently prioritise US-centric naming conventions as the baseline for identifiable names.

5.2.3 Transliteration and Script Representation

A further limitation arises from the transliteration pipeline used to standardise non-Latin scripts. The `anyascii` conversion process produces highly simplified Latin approximations, which in some cases remove phonemic information entirely. These transformations introduce an asymmetry in representation quality across language groups. As a result, part of the observed disparity for certain non-Latin-origin groups may be amplified by transliteration artefacts rather than reflecting purely lexical differences in spell-checker coverage. The reflections upon the solely Latin-script origin names provides a clean analysis unaffected by these constraints.

5.2.4 Methodological Scope

Several additional considerations affect how these results should be interpreted in relation to prior work and real-world systems.

First, the analysis is based on specific versions of each spell-checking tool at the time of execution. As dictionaries and rule sets are continuously updated, particularly for proper noun handling and multilingual support, the results represent a snapshot of system behaviour rather than a fixed property of the tools. This means that absolute values may shift over time, even if the underlying structural patterns persist.

Additionally, the pipeline is fully automated and does not include human validation, so the analysis cannot distinguish between culturally rare names and names that are widely used but missing from dictionaries. A targeted human evaluation would be required to separate these cases and refine the interpretation of recognition failures.

5.3 Future Work

While this study provides evidence of systematic disparities in spell-checking behaviour across cultural and linguistic name groups, several directions remain for extending and strengthening this work.

A key next step is the incorporation of human-centred evaluation. As this analysis is entirely computational, it cannot distinguish between names that are genuinely rare and those that are absent due to incomplete or biased lexicons. A targeted user study involving participants from Global Majority naming backgrounds would help to ground these findings in lived experience, assessing not only whether corrections occur, but

how they are perceived, whether they are accepted or overridden, and whether repeated flagging produces cumulative user burden. This would directly respond to Blodgett et al. [21] critiques of representational bias research, which emphasise the importance of connecting computational measures to lived harm.

A second direction involves extending analysis to proprietary and widely used consumer-facing systems. The tools examined in this study are representative of common open or hybrid spell-checking architectures, but they do not include systems embedded in commercial productivity software such as Google Docs or Grammarly, nor mobile keyboard autocorrect systems, which are likely to be among the most frequently encountered by end users. Evaluating these systems would improve real-world applicability, although such work would require alternative data collection strategies such as interface simulation or controlled user interaction studies due to limited programmatic access.

Future work could also extend analyses to modern language systems based on large language models. Unlike dictionary-based spell-checkers, LLM-based systems generate corrections probabilistically from large-scale training data rather than explicit lexicons. This introduces a different mechanism through which bias may arise, potentially amplifying or reshaping the disparities observed in this study. Comparing traditional spell-checking systems with LLM-based correction models would help clarify whether the observed patterns are specific to lexicon-driven architectures or reflect broader properties of language modelling systems.

A further extension involves temporal analysis of system updates. Since spell-checking dictionaries and language models are periodically updated, analysis across versions would allow assessment of whether coverage of Global Majority naming conventions is improving, stagnating, or degrading over time.

Finally, improvements to transliteration and script handling would strengthen cross-script comparability. Replacing the current `anyascii` pipeline with more linguistically informed romanisation systems would better preserve phonemic structure. This would allow a more accurate separation of transliteration effects from true lexical coverage gaps, particularly for languages such as Arabic and Vietnamese where diacritics and vowel structure carry meaningful linguistic information.

Together, these directions highlight the need to move beyond purely computational auditing toward more grounded, system-level, and user-centred evaluation of spelling correction technologies.

5.4 Final Remarks

This project set out to make measurable something that typically operates silently in the background of digital communication: the differential treatment of personal names from different cultural origins within spell-checking systems. These systems are widely perceived as neutral infrastructure, yet the findings of this study indicate that they encode systematic disparities in how names are recognised and transformed.

The “red underline” is not a neutral technical event. For a child completing schoolwork, a job applicant drafting a CV, or a patient filling in a medical form, repeated signalling

that their name is incorrect constitutes a small but cumulative harm. While each instance is individually minor, together these interactions shape how linguistic legitimacy is experienced in everyday digital environments.

Across all tools and analytical approaches, the results suggest that these disparities are not isolated artefacts of specific implementations, but reflect structural properties of spell-checking systems. The persistence of gaps across script conditions, tool architectures, and controlled specifications indicates systematic privileging of naming conventions more closely aligned with Global North and European linguistic forms.

At the same time, the limitations of this study are substantial and should be foregrounded. The analysis is entirely computational and does not include direct participation from affected communities. The categories used are necessarily imperfect proxies for cultural and linguistic identity, and the dataset reflects the uneven global reach of a single platform rather than a fully representative global population. These constraints mean the findings should be interpreted as evidence of systematic patterns rather than definitive measures of harm.

Despite these limitations, the results contribute to a broader accountability agenda by making such disparities visible and measurable. By rendering these patterns legible at scale, this work provides a basis for further scrutiny by affected communities, developers, and policymakers.

If language technologies are to be truly inclusive, the lexical representation of personal names must be treated as a core system requirement rather than a dictionary edge case. By making these systematic disparities visible, this study supports the broader argument that equitable name recognition in spell-checkers is a primary design requirement of language technology, not a secondary concern [21].

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Appendix A

Appendix

A.1 Evaluation of Enrichment

Table A.1: Distribution of `langdetect` Predictions by Confidence Level and Relative Overrepresentation

Language	**High-Confidence (%)** ($P \geq 0.7$)	**Low-Confidence (%)** ($P < 0.7$)	**Ratio (Low / High)**
English	4.91	6.24	1.27
Swahili (macrolanguage)	5.80	6.67	1.15
German	5.24	5.95	1.14
Indonesian	6.42	7.21	1.12
Turkish	3.40	3.79	1.11
Tagalog	4.79	5.18	1.08
Somali	5.59	5.42	0.97
Arabic	3.71	0.00	0.00
Italian	3.36	0.00	0.00
Korean	5.45	0.00	0.00

Shares represent the proportion of each predicted language within all outputs at the given confidence level. The overrepresentation ratio compares low- to high-confidence shares. Values > 1 indicate that a language is disproportionately predicted when the model is uncertain, suggesting it may function as a default or fallback prediction.

Table A.2: Top 10 Represented Countries in the Cleaned Dataset

Country	**Count** (n)	**Percentage** (%)
Saudi Arabia	25,196	4.41
United States	25,106	4.40
South Africa	20,677	3.62
Malaysia	19,911	3.49
Russian Federation	16,633	2.91
Italy	15,697	2.75
Egypt	15,459	2.71
Nigeria	15,298	2.68
Brazil	13,753	2.41
Korea, Republic of	13,751	2.41

Note: Percentages are calculated based on the total number of cleaned names in the dataset. Geographic distribution reflects the inferred primary country of origin for each name entry as determined by the prediction model.

A.2 Recognition Rate Exploration

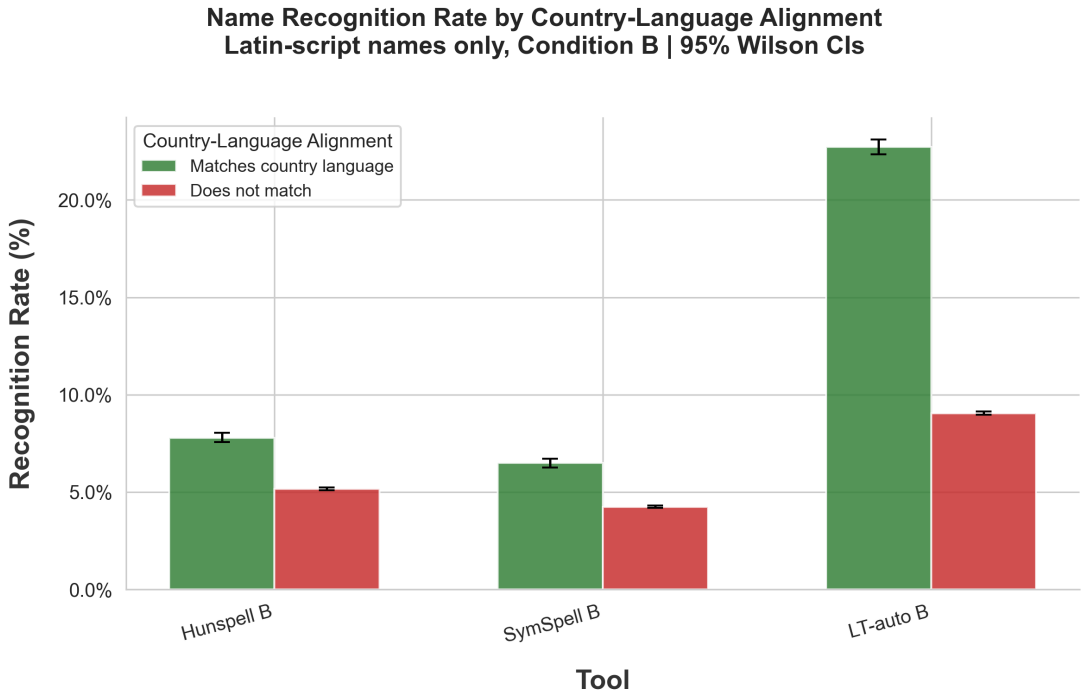

Figure A.1: Comparing recognition rates when langdetect's language prediction does vs does not match the dataset top_country prediction

A.3 Correction Behaviour Analysis

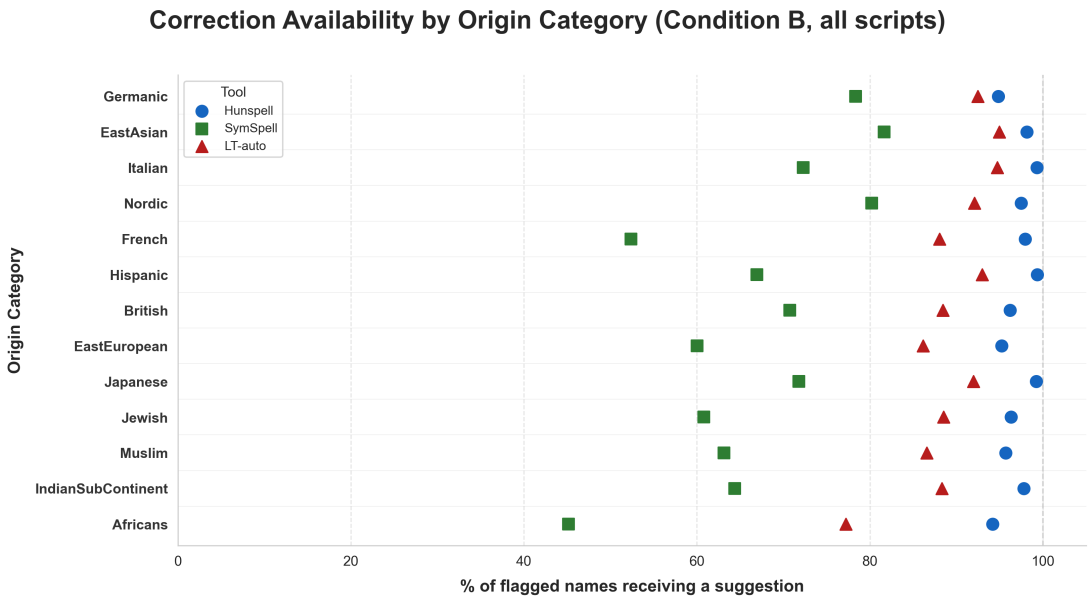

Figure A.2: Each tool's rate of corrections, grouped by ethnicolr category

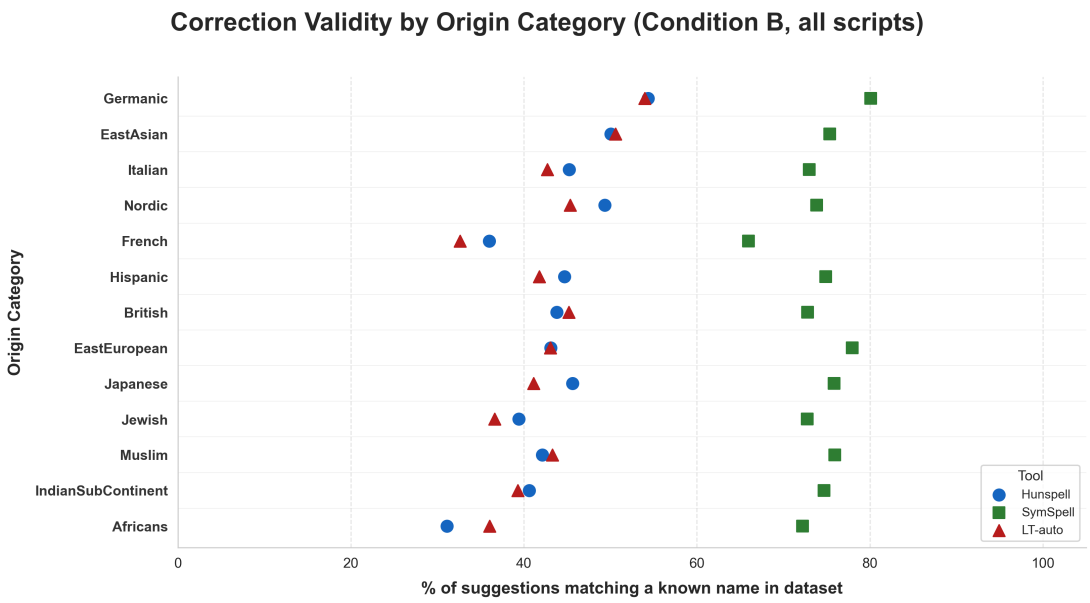

Figure A.3

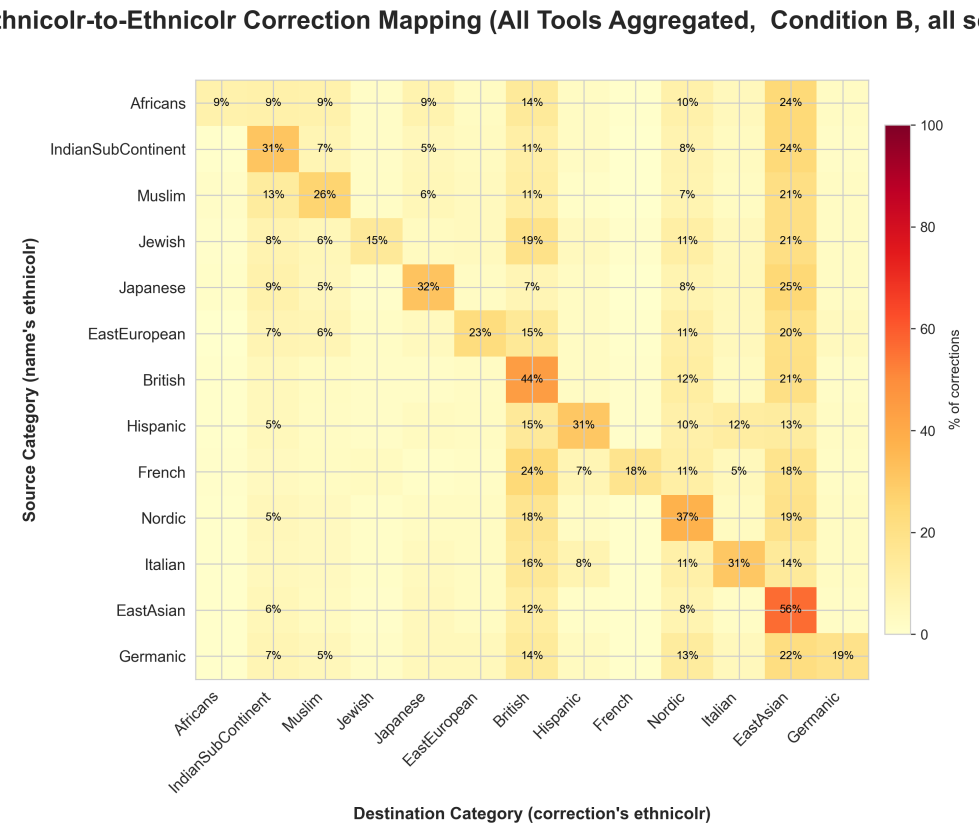

Figure A.4: Each tool's rate of corrections appearing in dataset, grouped by ethnicolr category